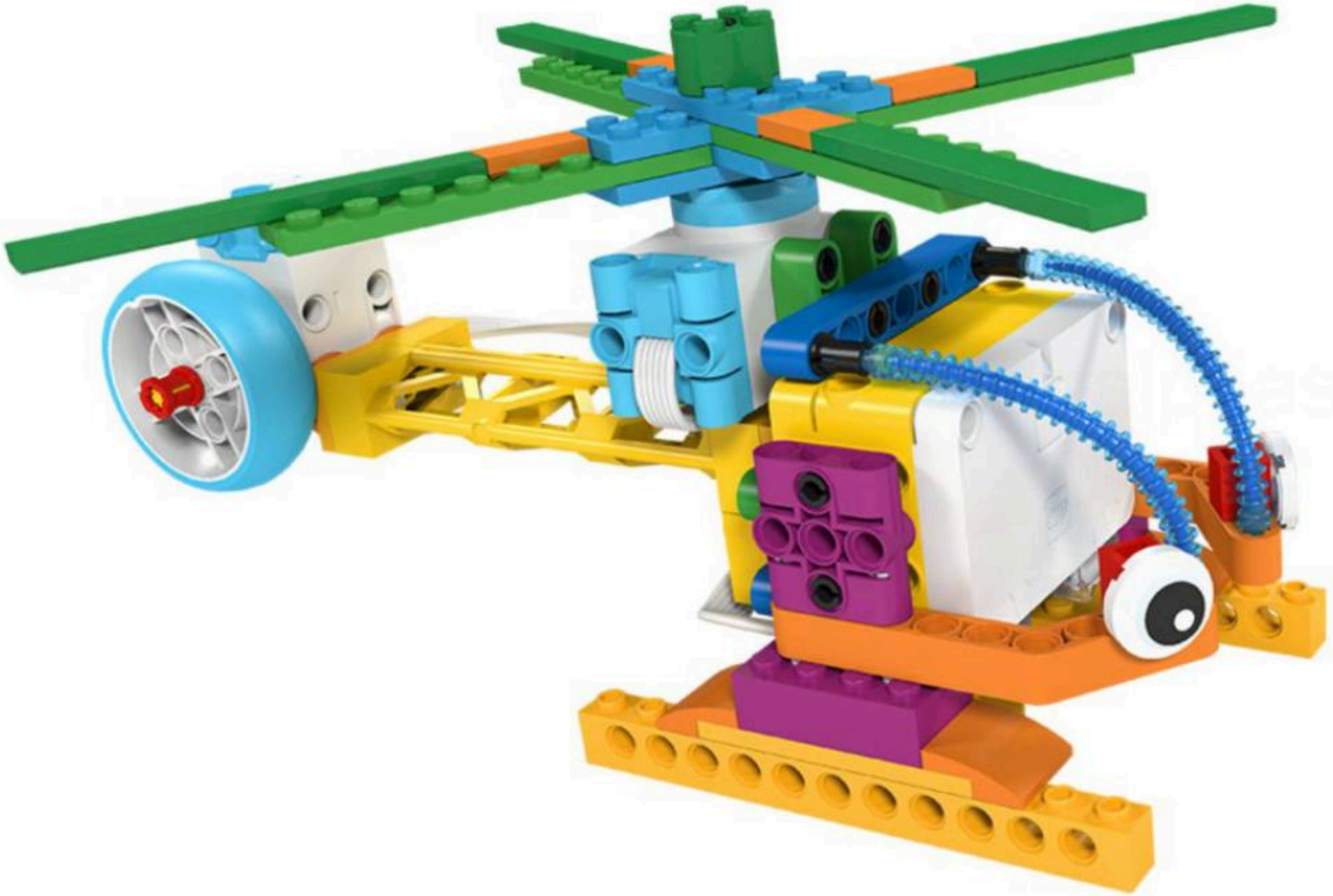


I gave this data to my friends from the university to decipher and create a forecast for the future.



When my friends found out that I started research related to the climate, they too began to think more about ecology and the state of the world around them



For example, Ashley is very concerned about supervolcanoes. She says that the eruption of such a volcano could almost destroy all of humanity!



Super volcanoes

I think we need to find out what she is talking about and whether it is really that dangerous and can affect people's lives





Hello! Well, supervolcanoes really did have a big impact on how our planet looks now



In fact, the term itself is not very scientific, but it describes the meaning very well. Supervolcanoes are really powerful volcanoes, the eruptions of which are very powerful, but not very frequent



Scientists believe that the last such eruption occurred about 70 thousand years ago. Then there was an eruption of the Toba volcano on the island of Sumatra.



There is research that climate change has led to a 'volcanic winter' - the phenomenon of a decrease in air temperature due to the huge amount of ash and sulfur gas in the air that blocks out sunlight



According to scientists' estimates, only 10-15 thousand people remained on Earth because of this eruption.



Such huge volcanoes form calderas - depressions that arise due to the collapse of the top of the volcano.



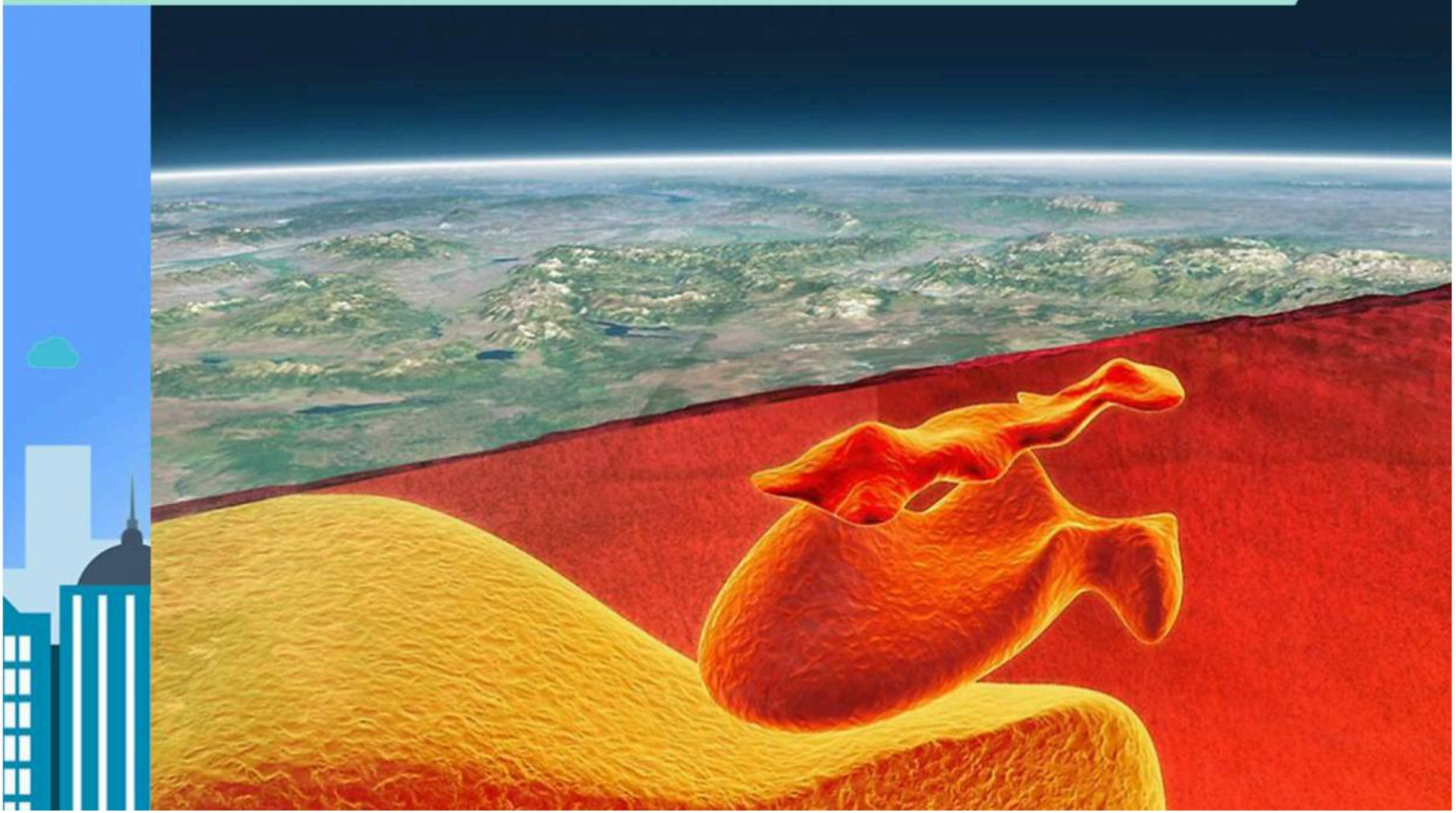
Calderas can be tens of kilometers in size. For example, the caldera of the supervolcano in the Yellowstone National Park in the USA has dimensions of 50x70 km



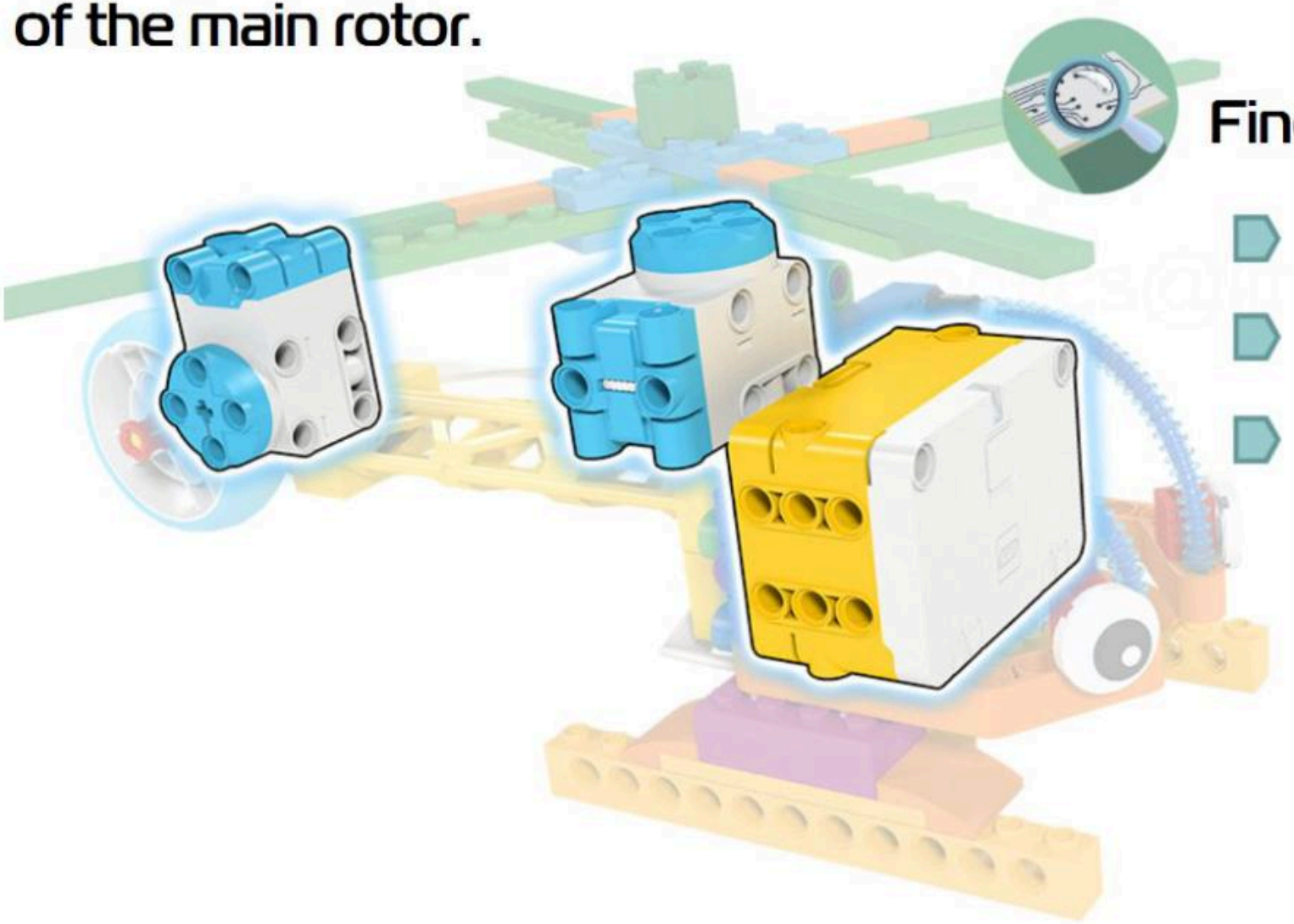
This supervolcano erupts every 600 thousand years and 640 thousand years have already passed since the last eruption. Thus, the condition of the caldera must be constantly monitored



The rise of the surface of the caldera indicates the accumulation of magma under it. We will be able to observe this process and the eruption will not be a surprise for us



The two robot's motors are used to drive the Main Rotor and the Tail Rotor. In such a design of the body, the Tail Rotor is needed to compensate for the reactional torque produced by the spinning of the main rotor.



Find:

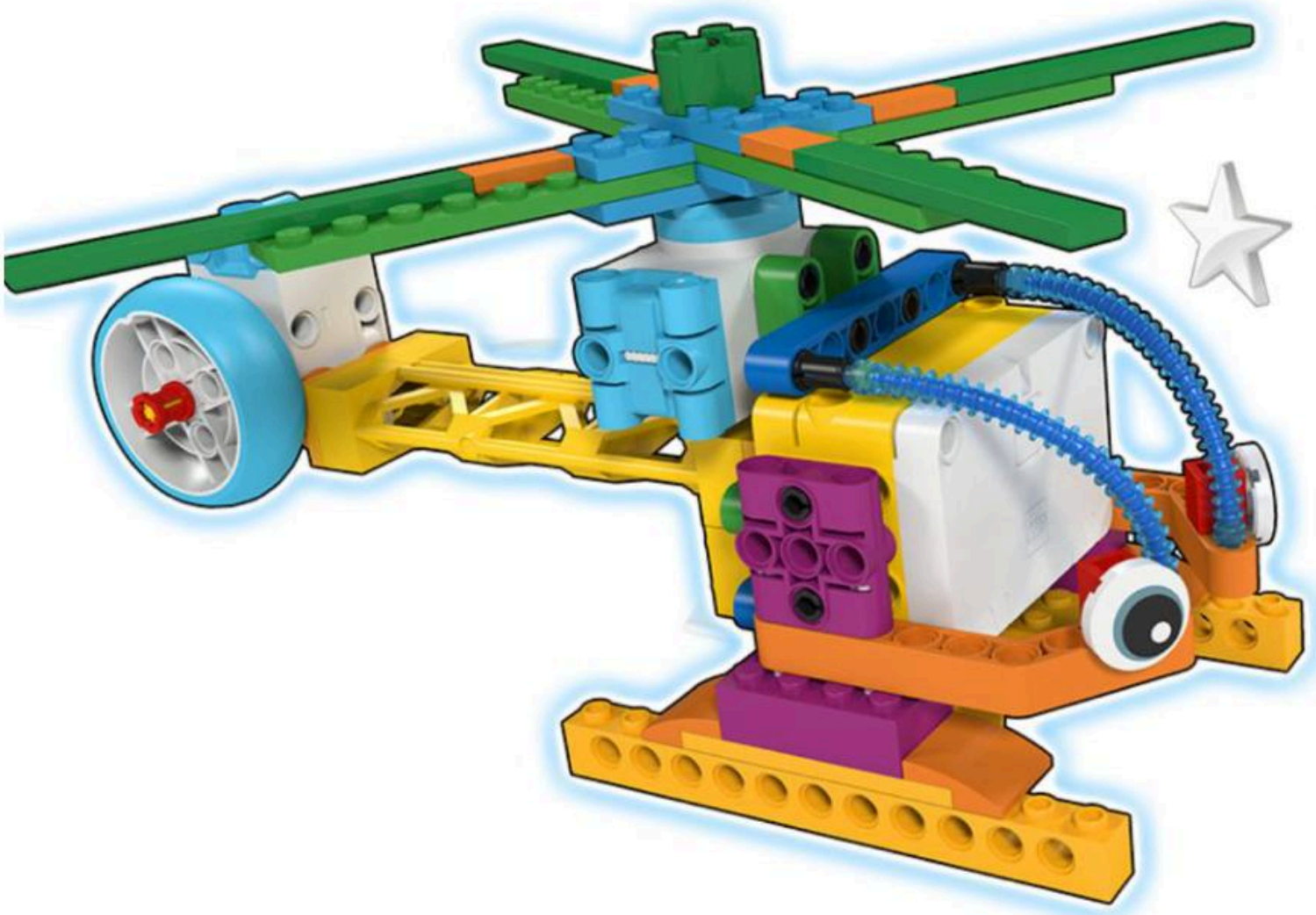
- Smarthub
- tail rotor's motor
- main rotor's motor

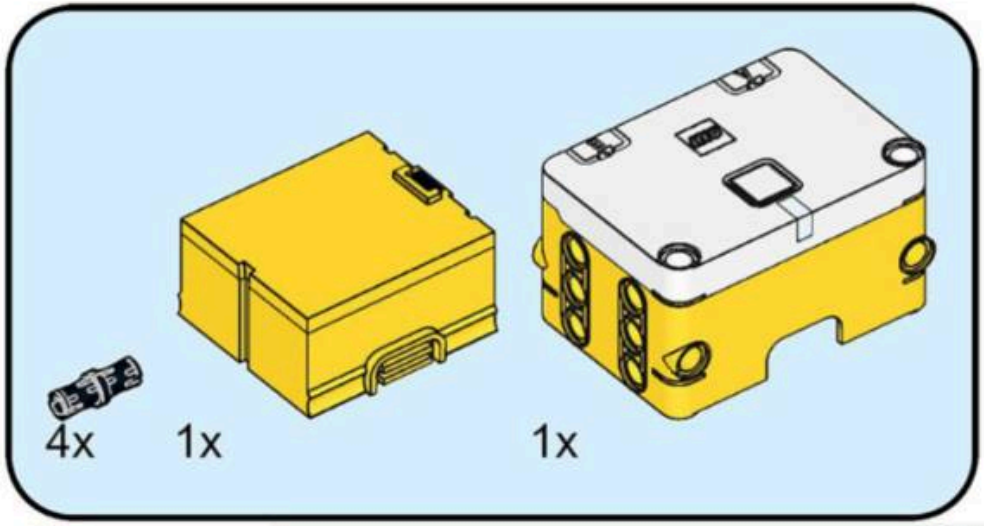


The robot has two screws. The main, propeller is responsible for the movement of the helicopter. At the same time, the rear or tail rotor is responsible for changing the direction of movement and compensates for the rotation of the helicopter caused by the main rotor.

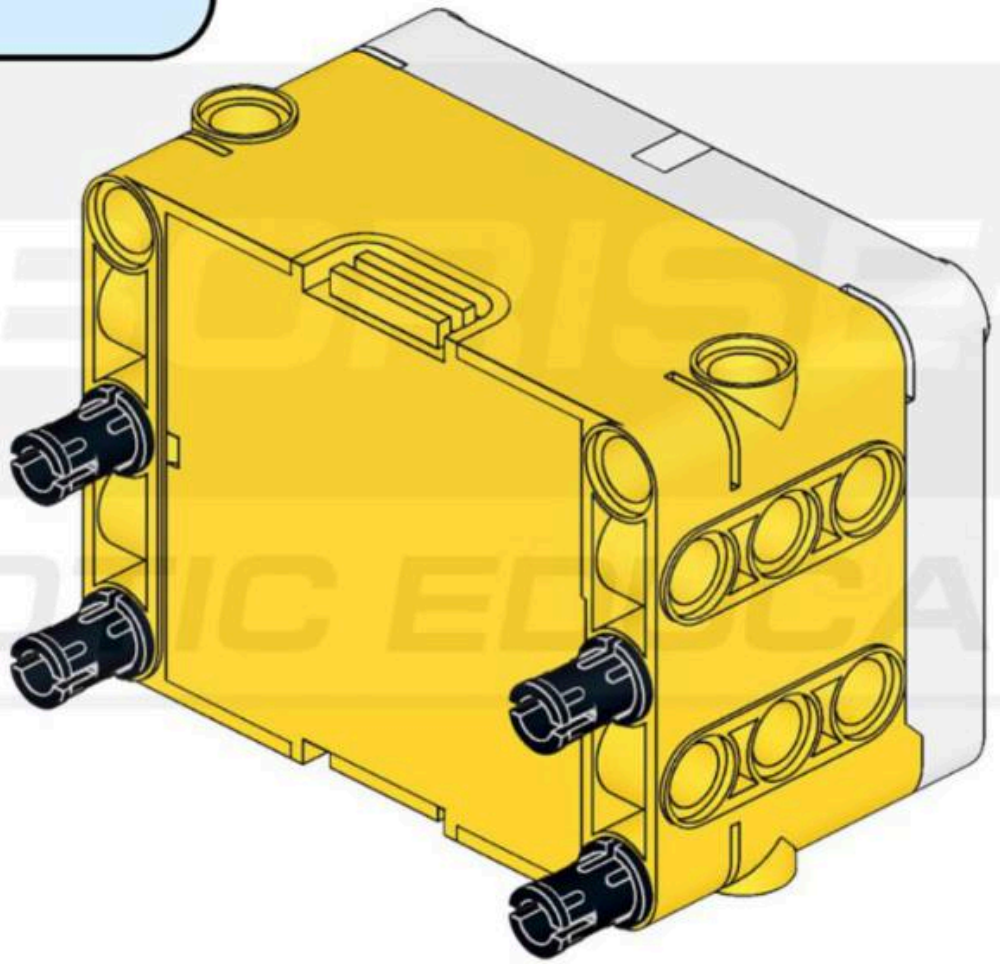


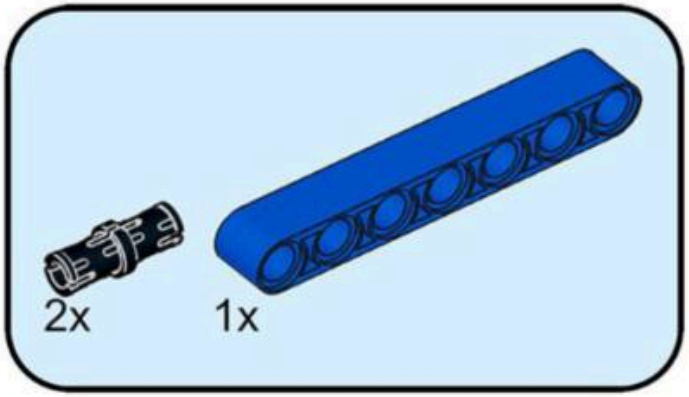
As a helicopter, the robot has all the main parts: main rotor, tail rotor, tail boom, cabin, and landing skids. Find all these parts of the robot's body below.



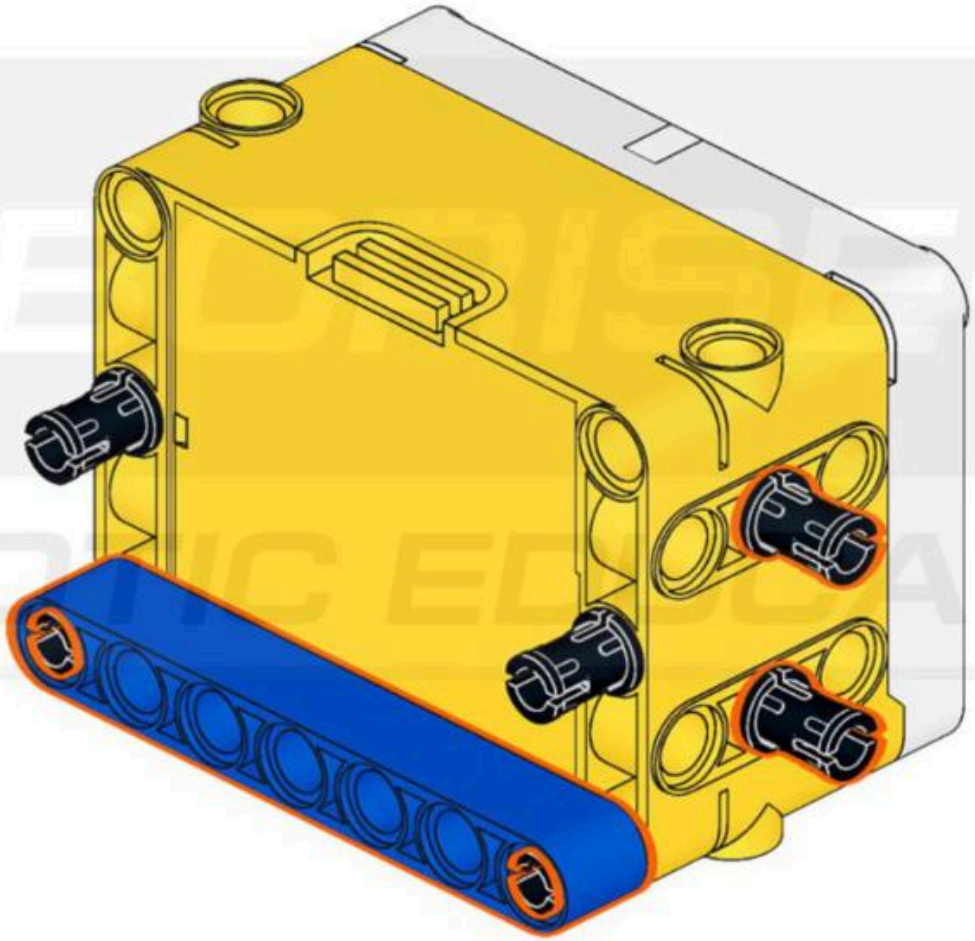


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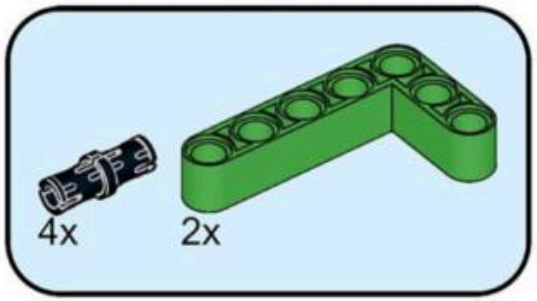




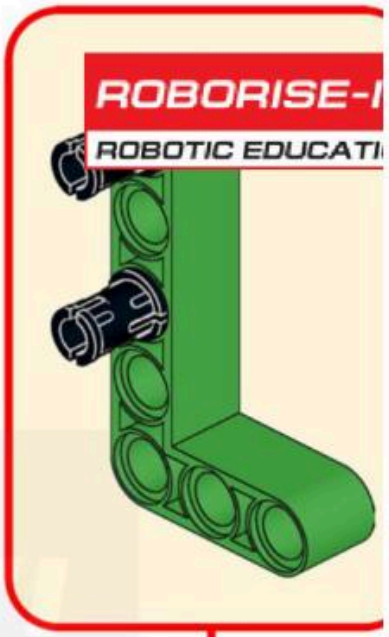
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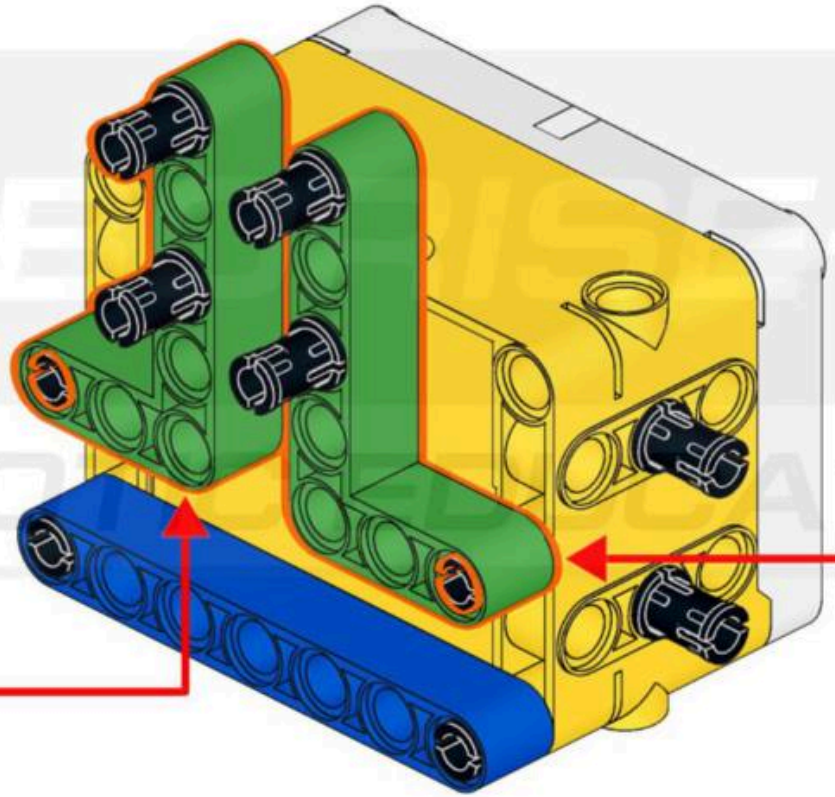


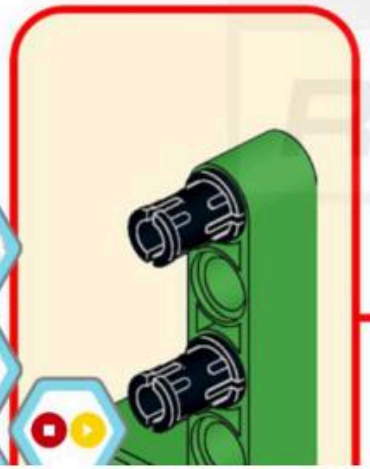
robotics@thebrainlab.com.au



3



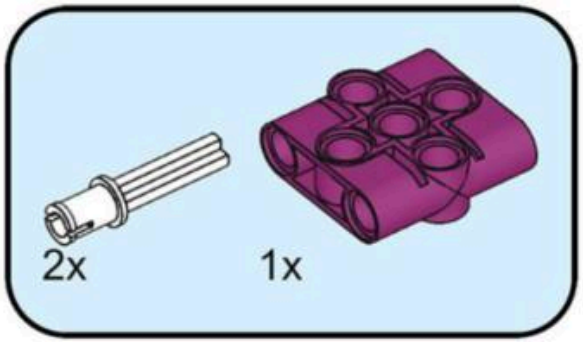


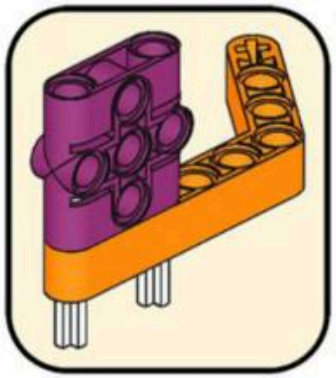


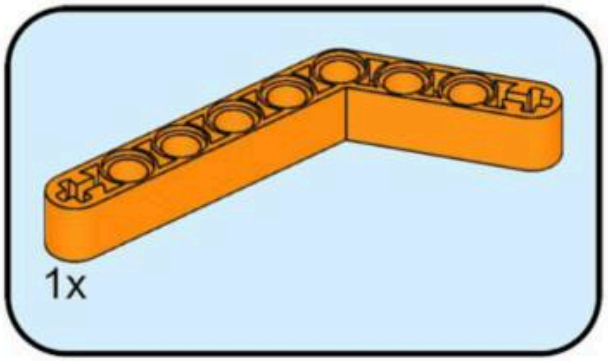




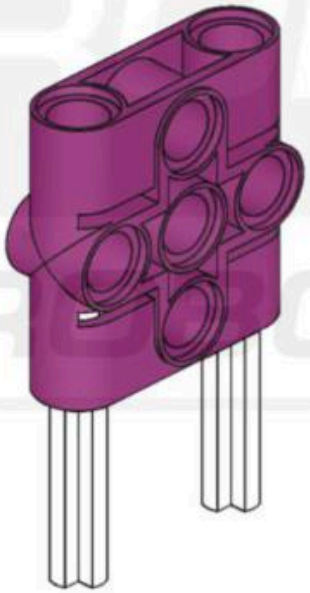
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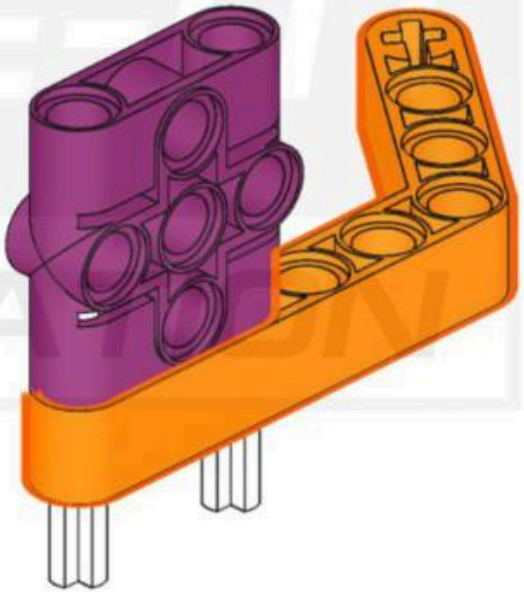




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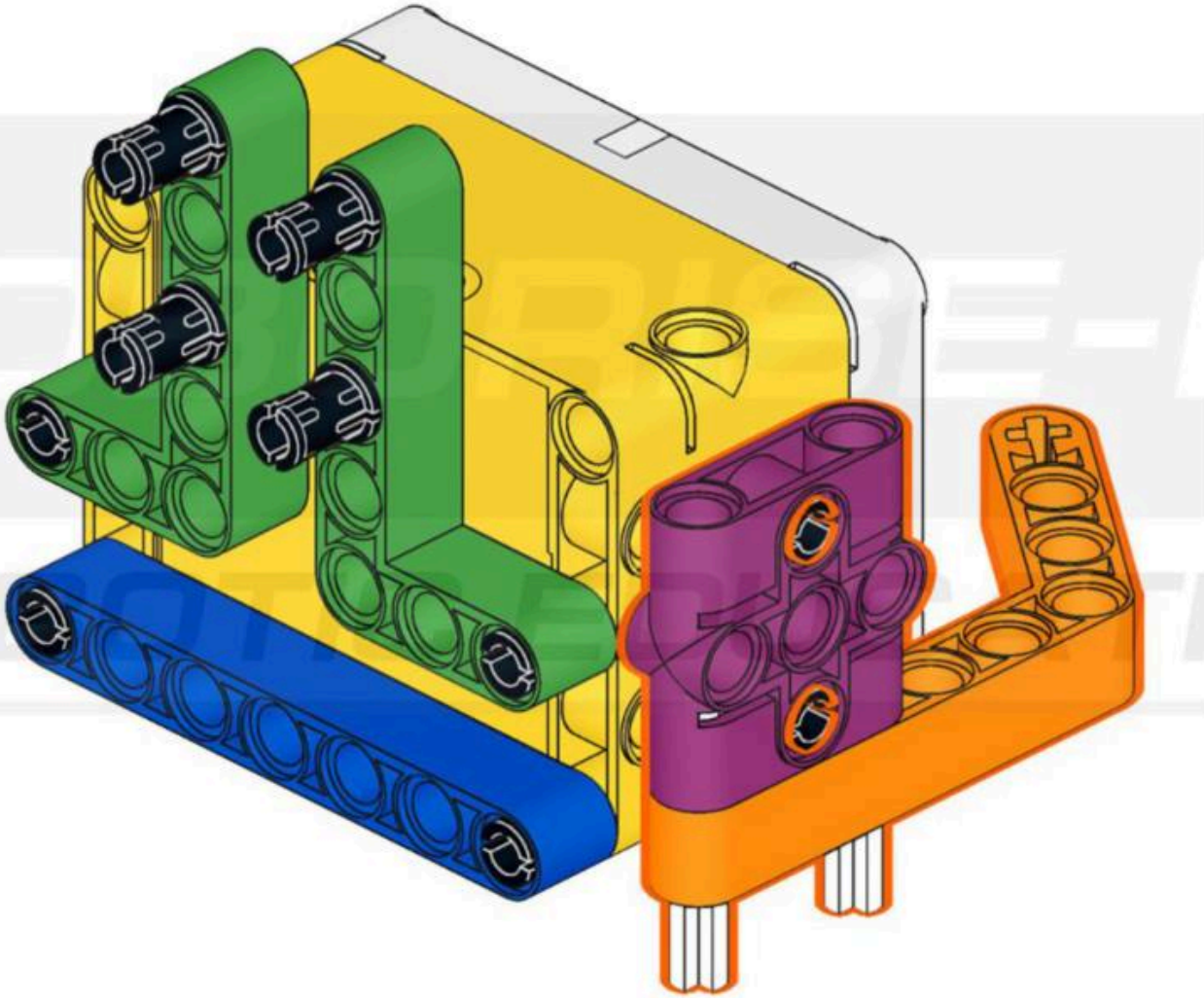


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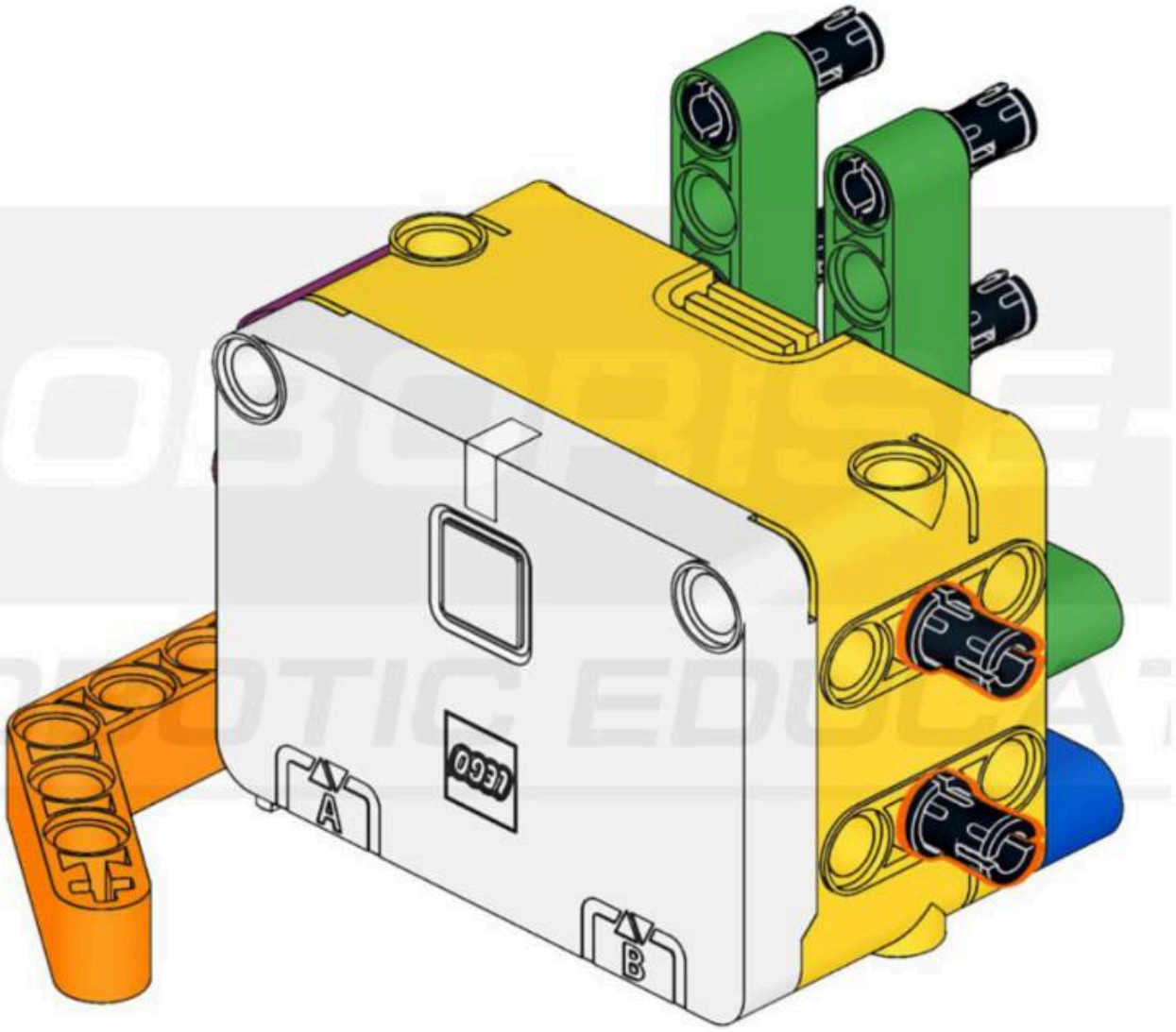
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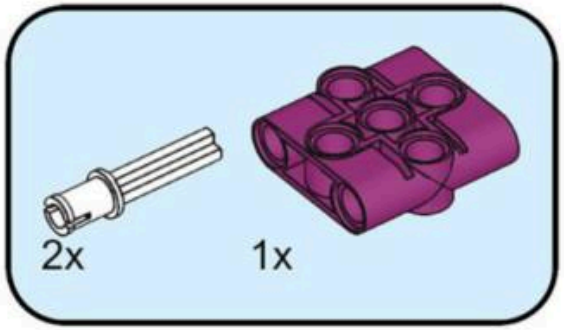


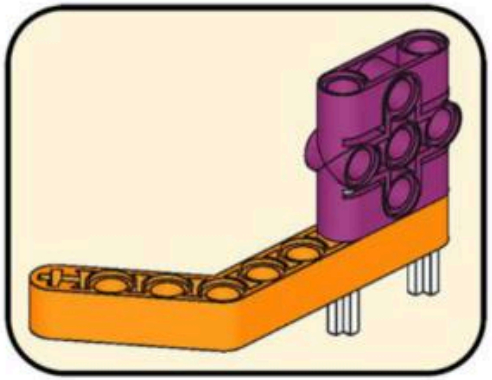


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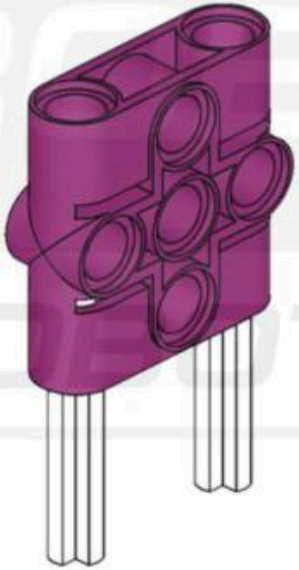


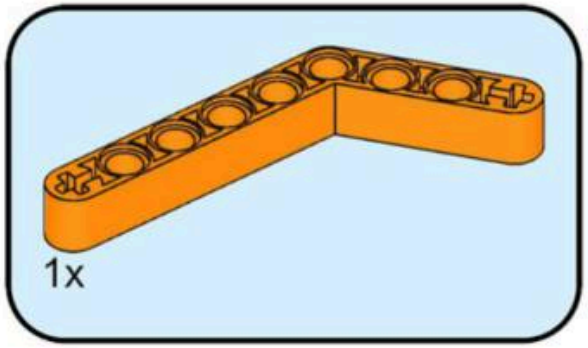
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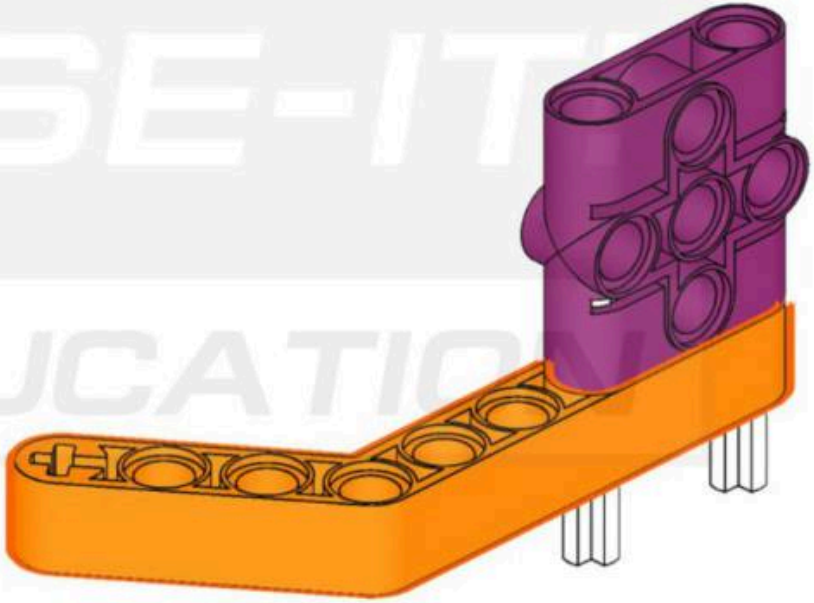


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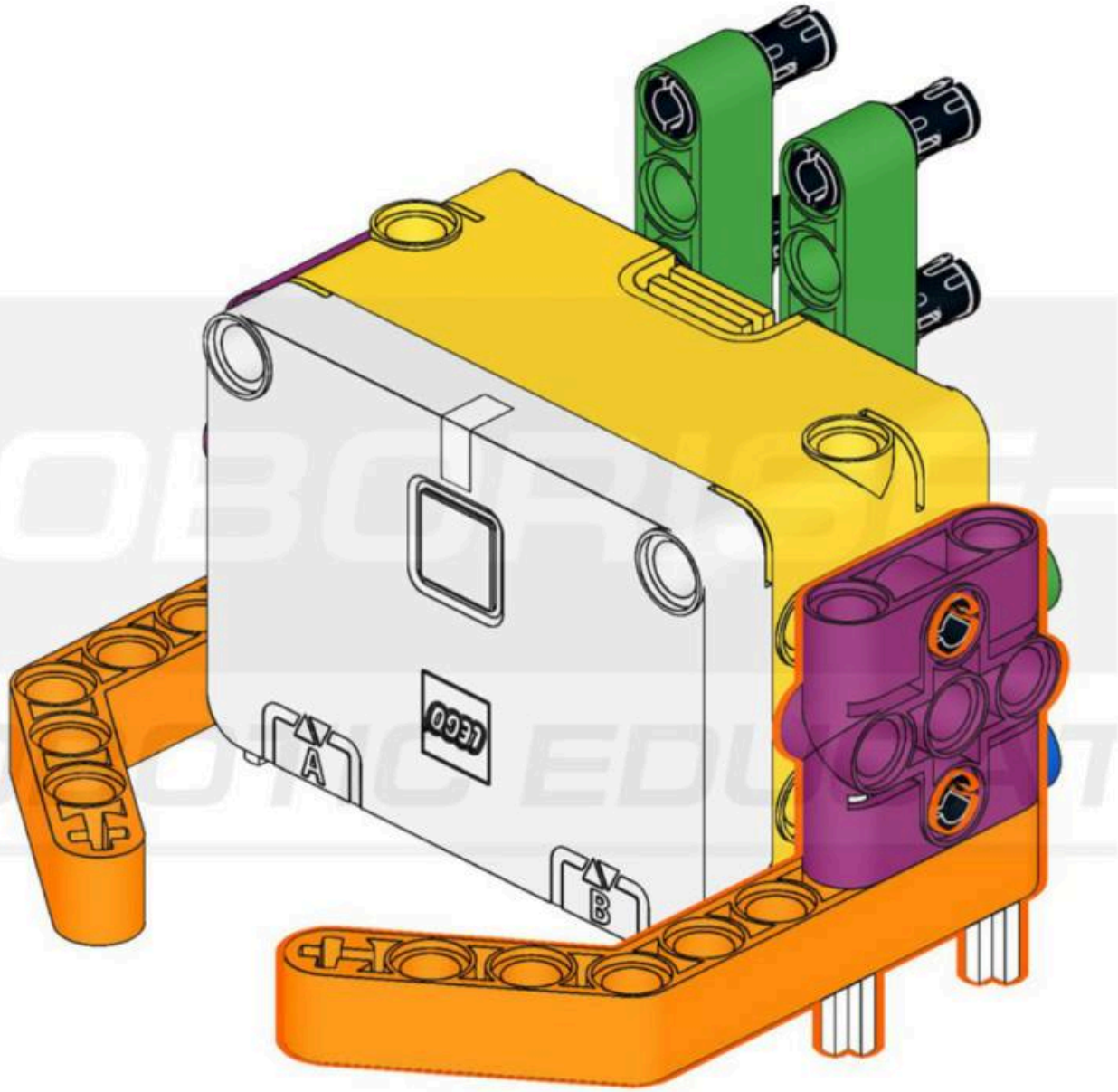


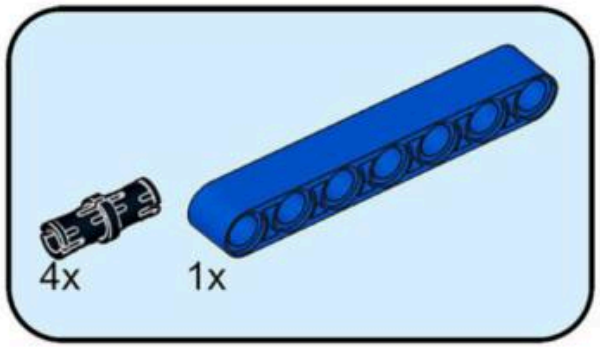
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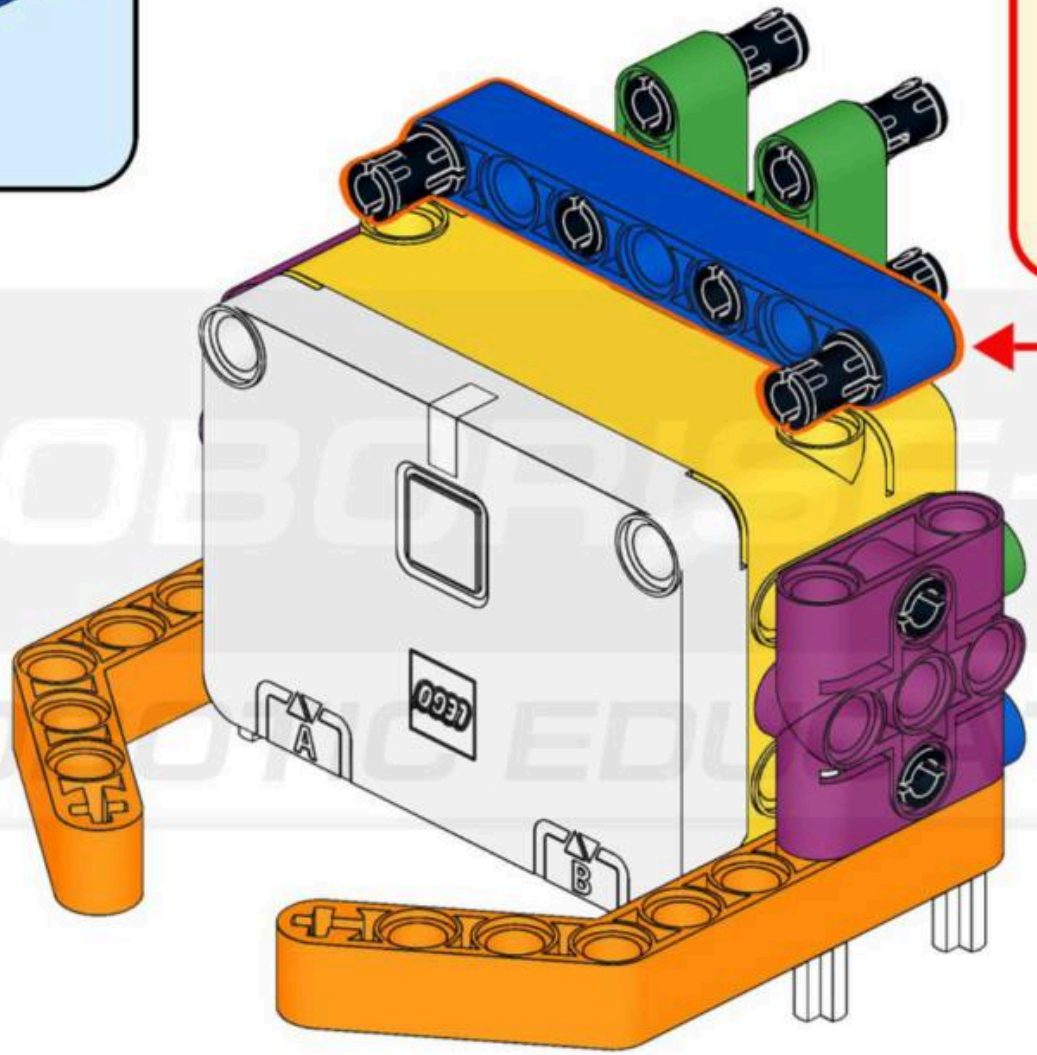
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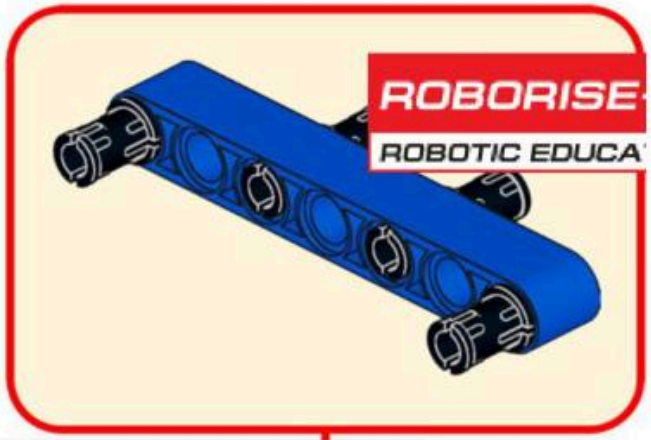
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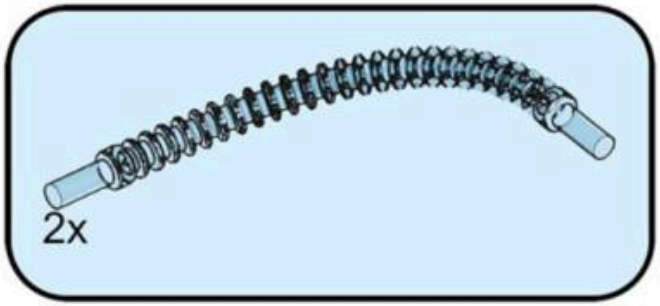


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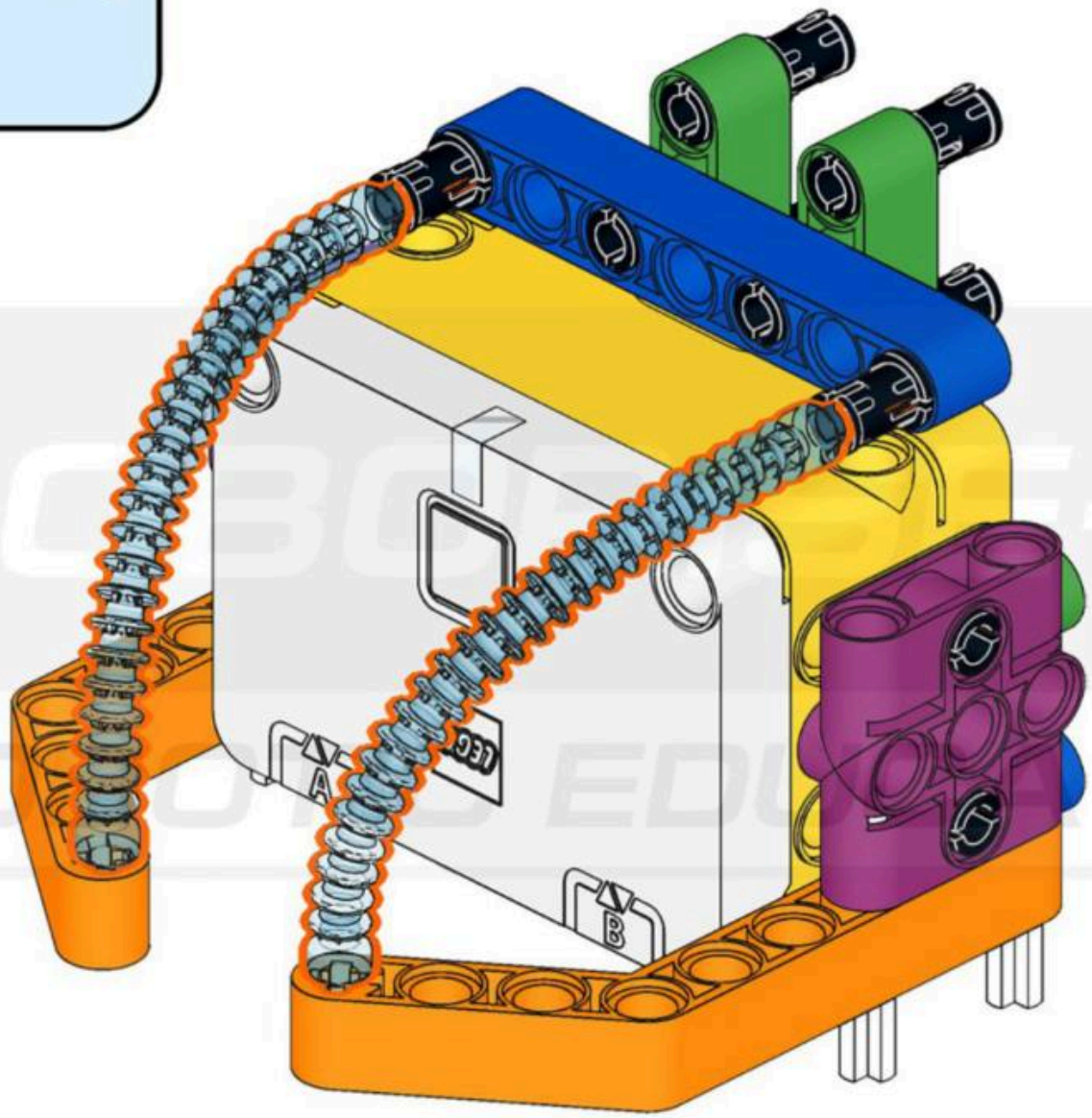


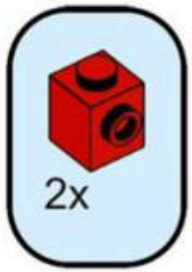


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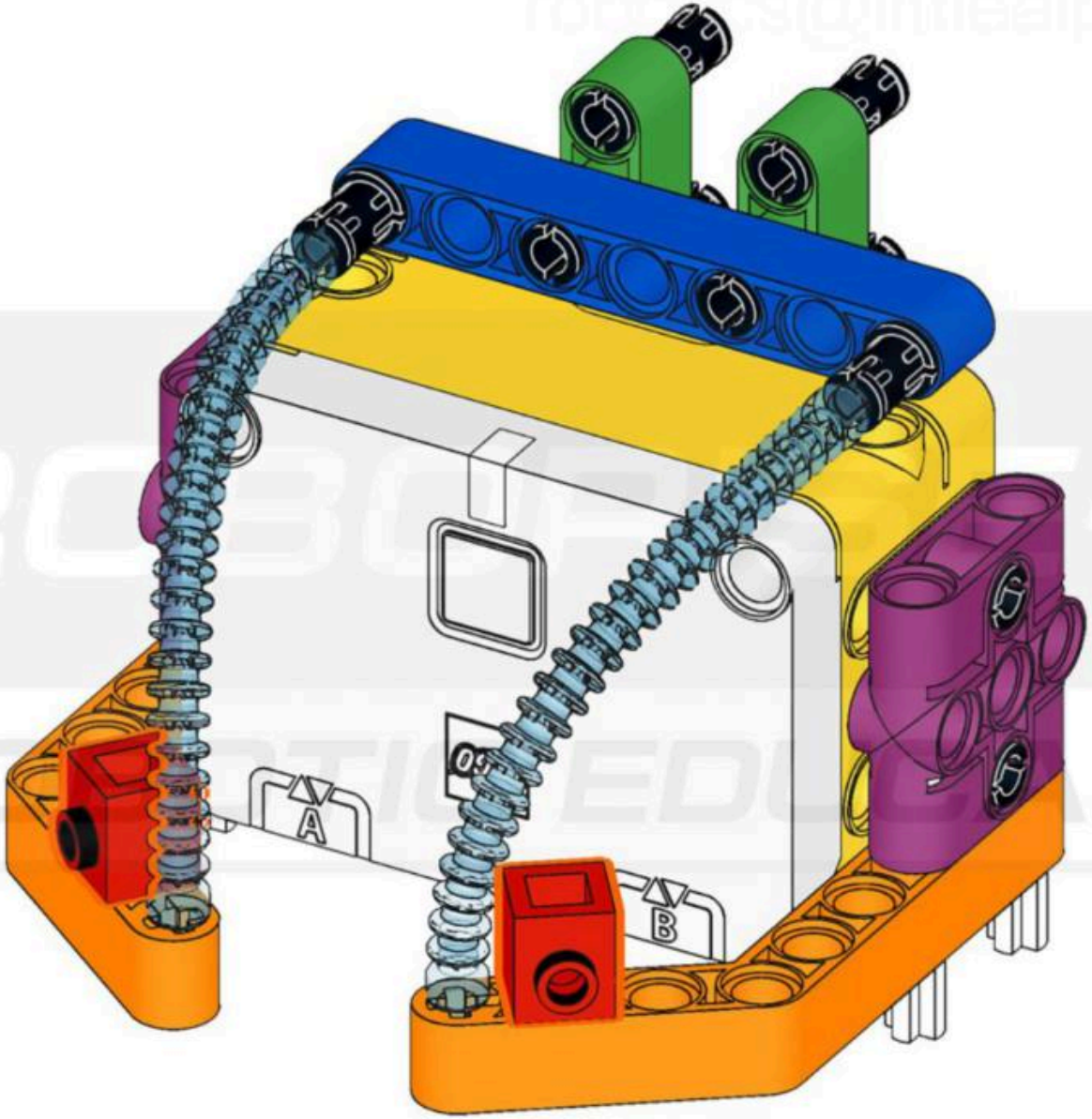


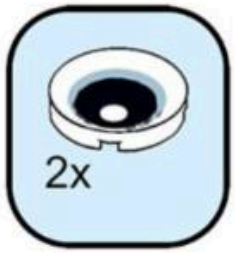
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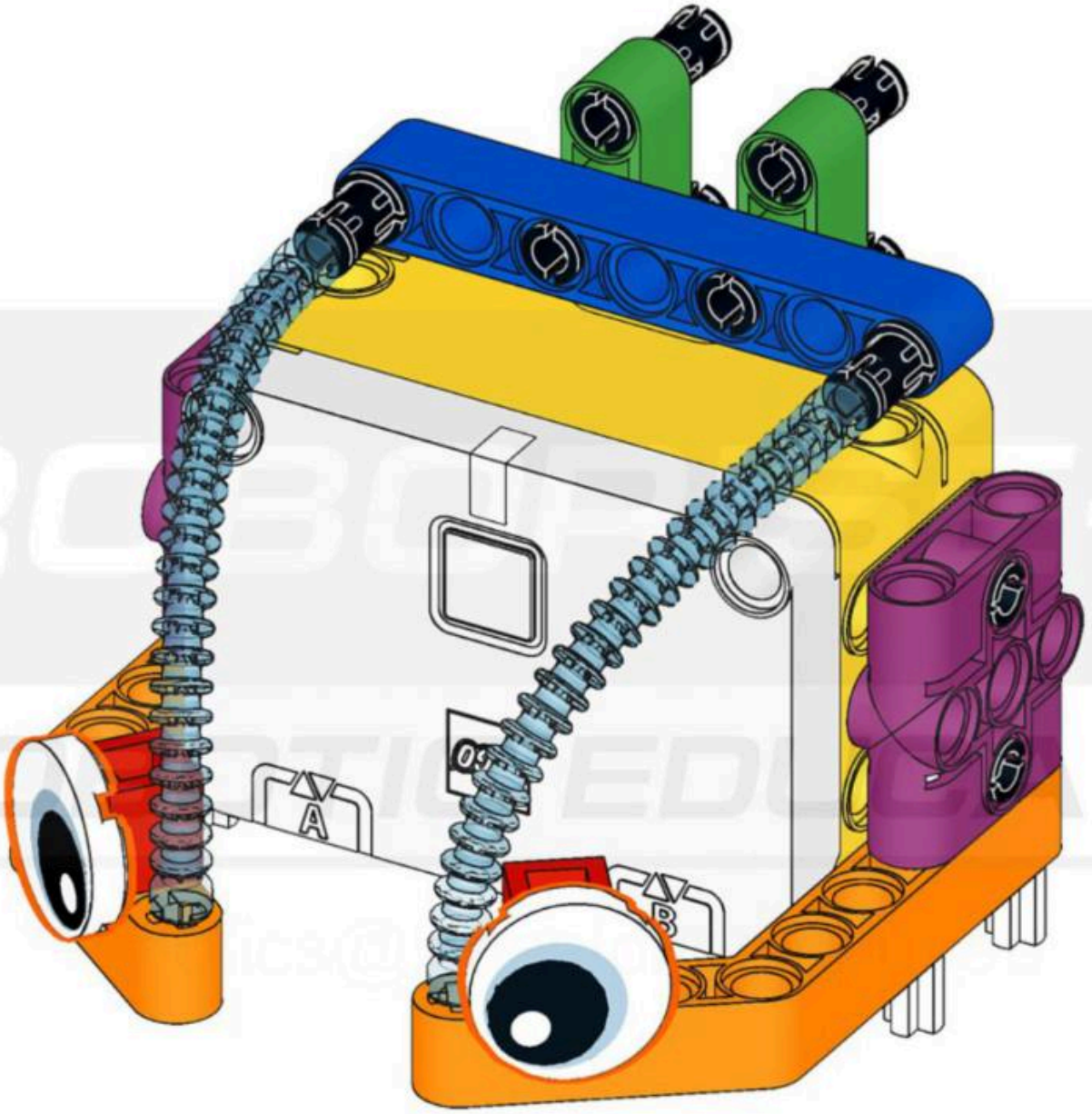


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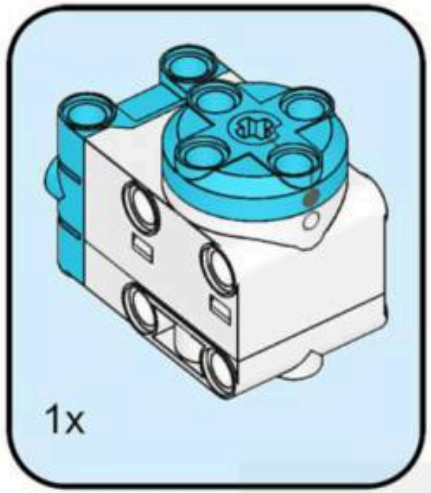




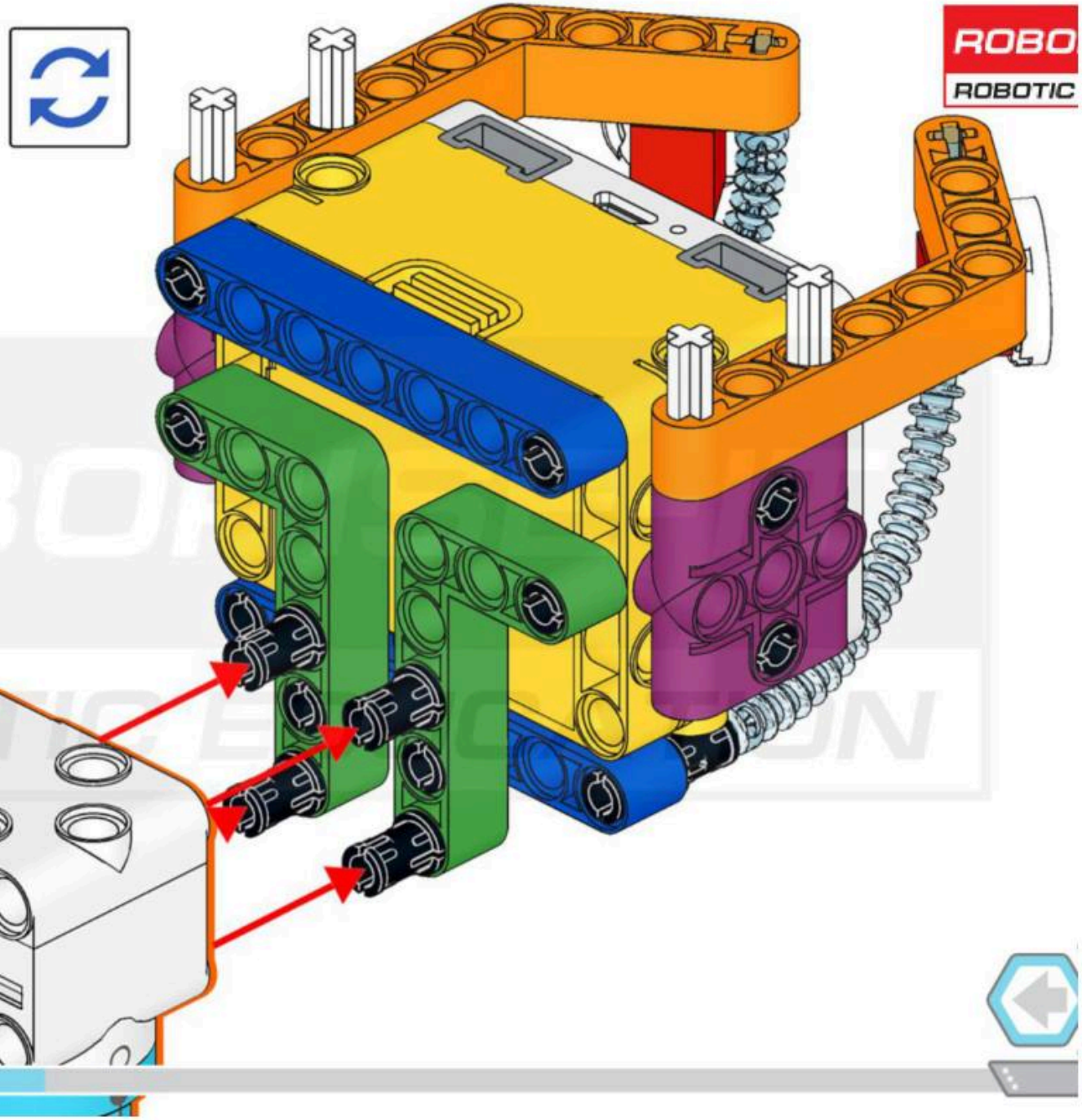
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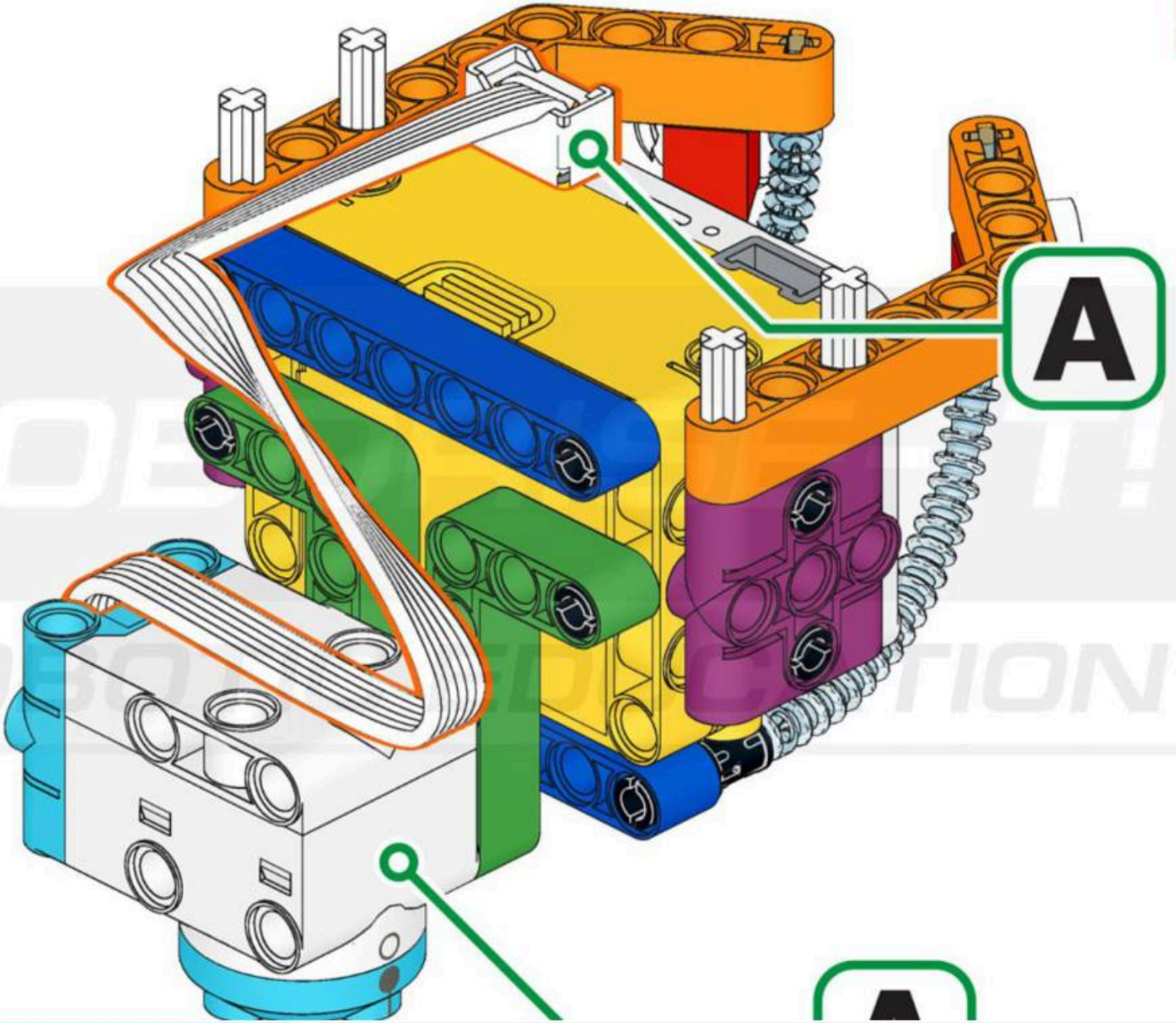




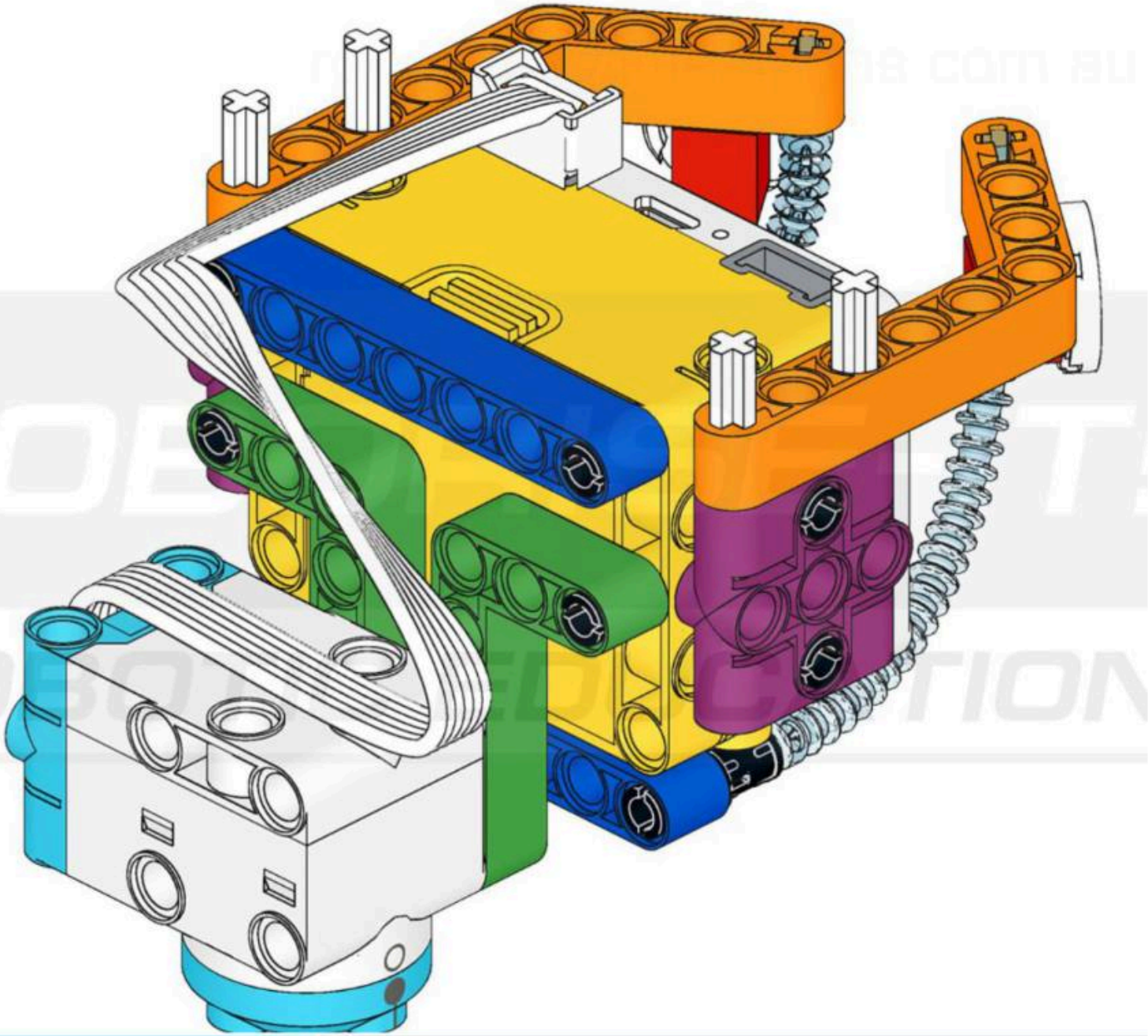
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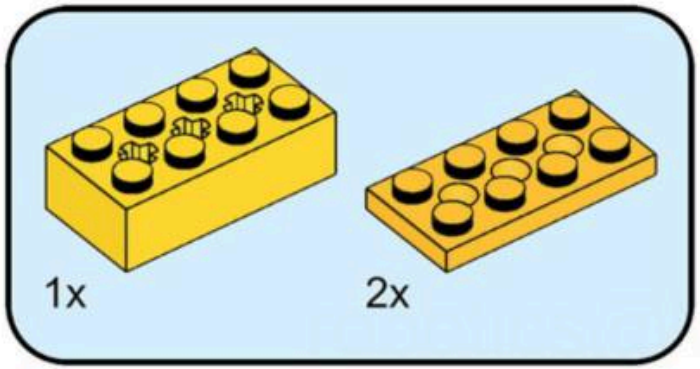


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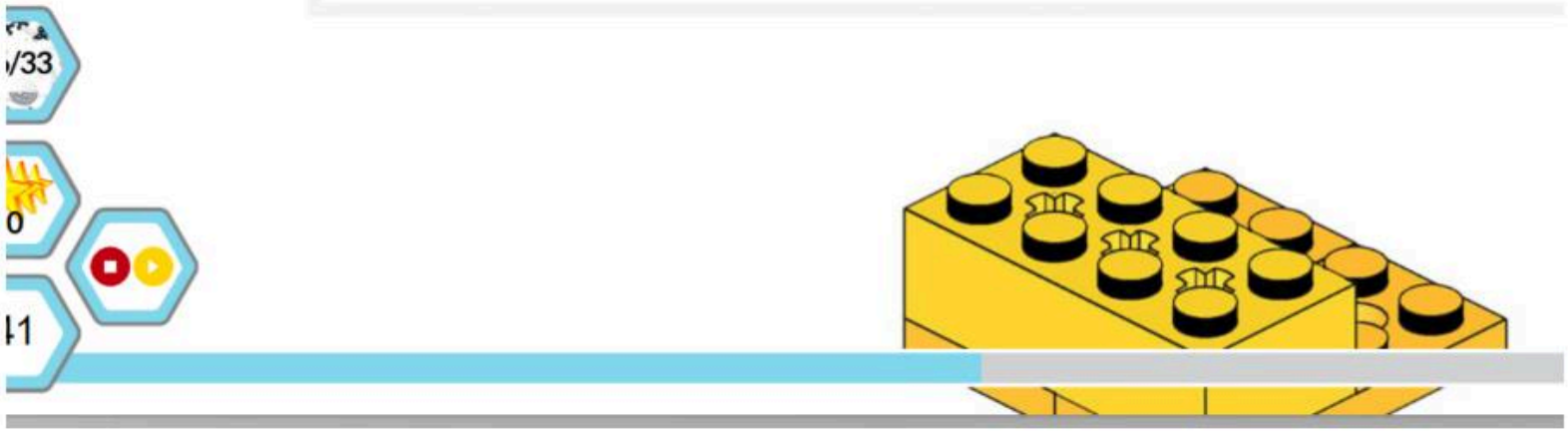
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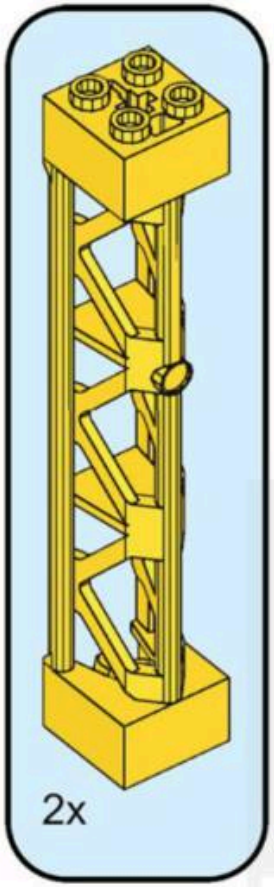




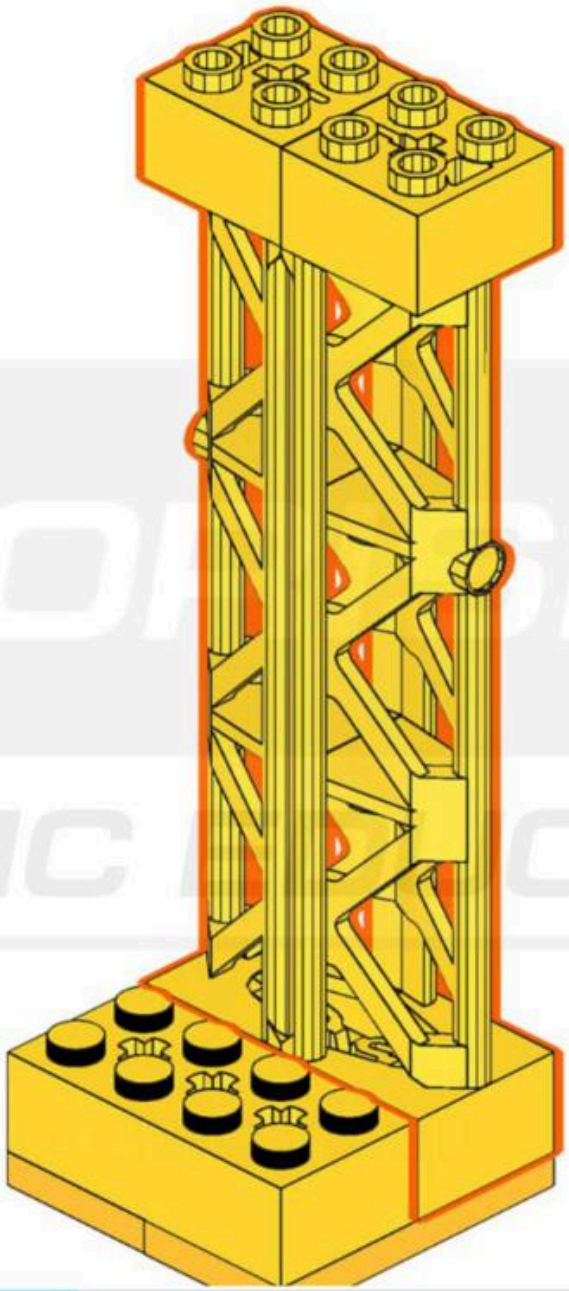
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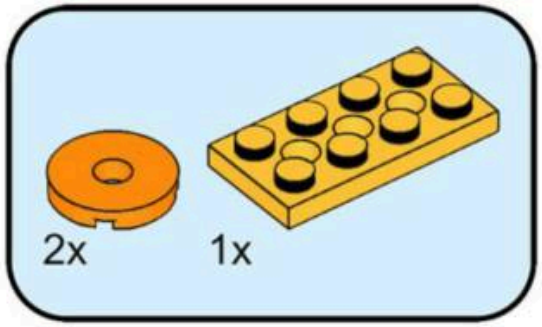
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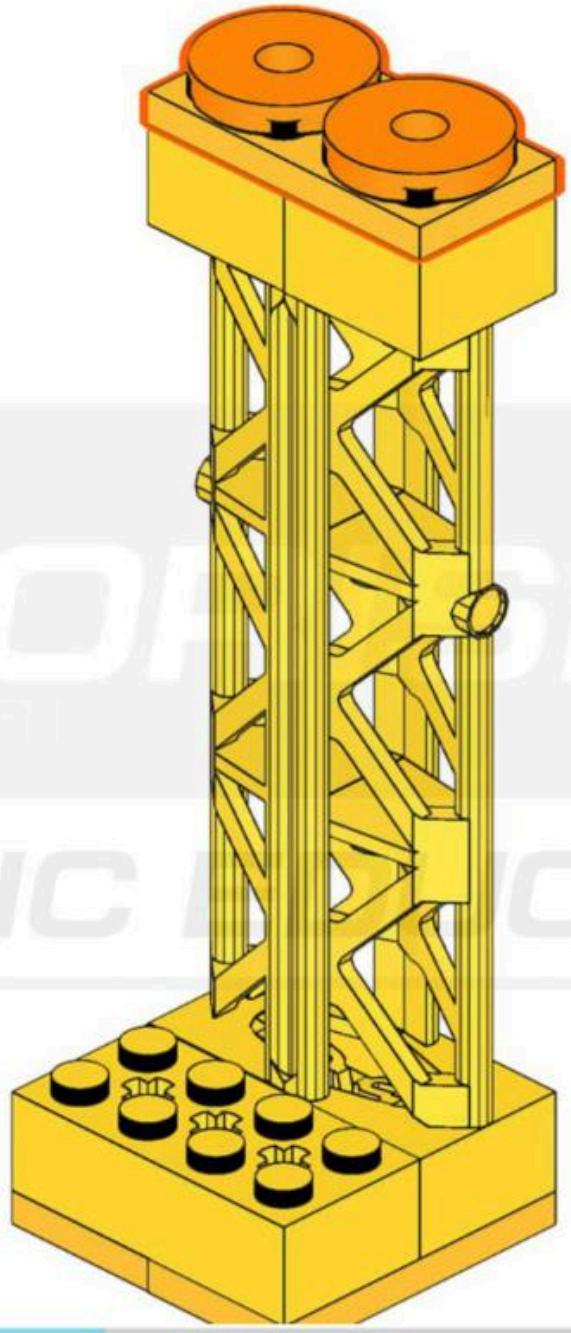


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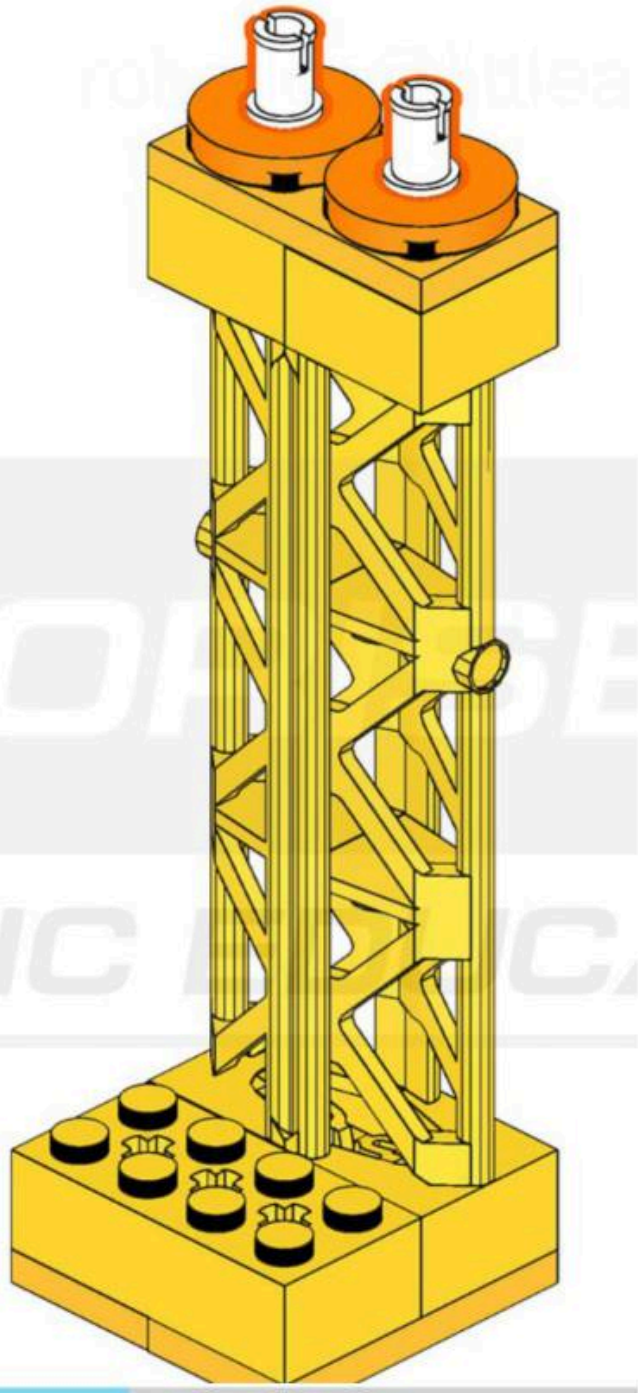
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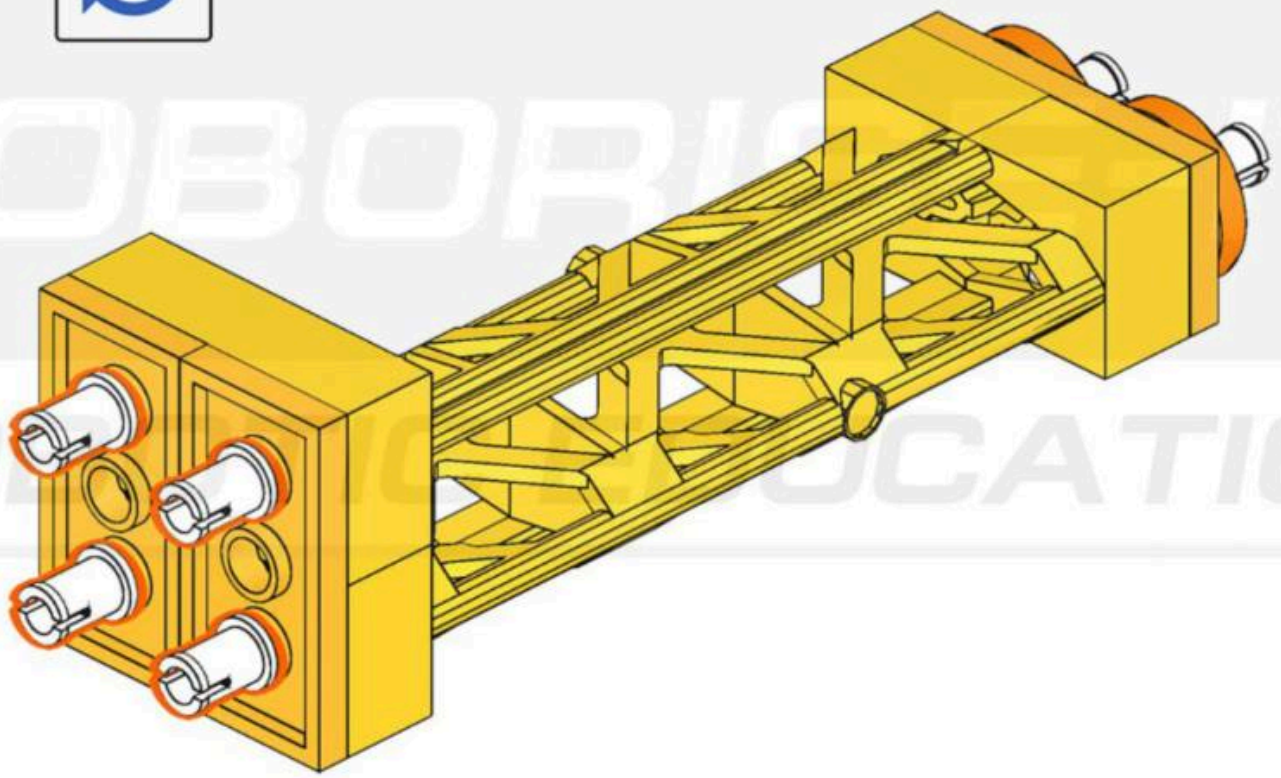
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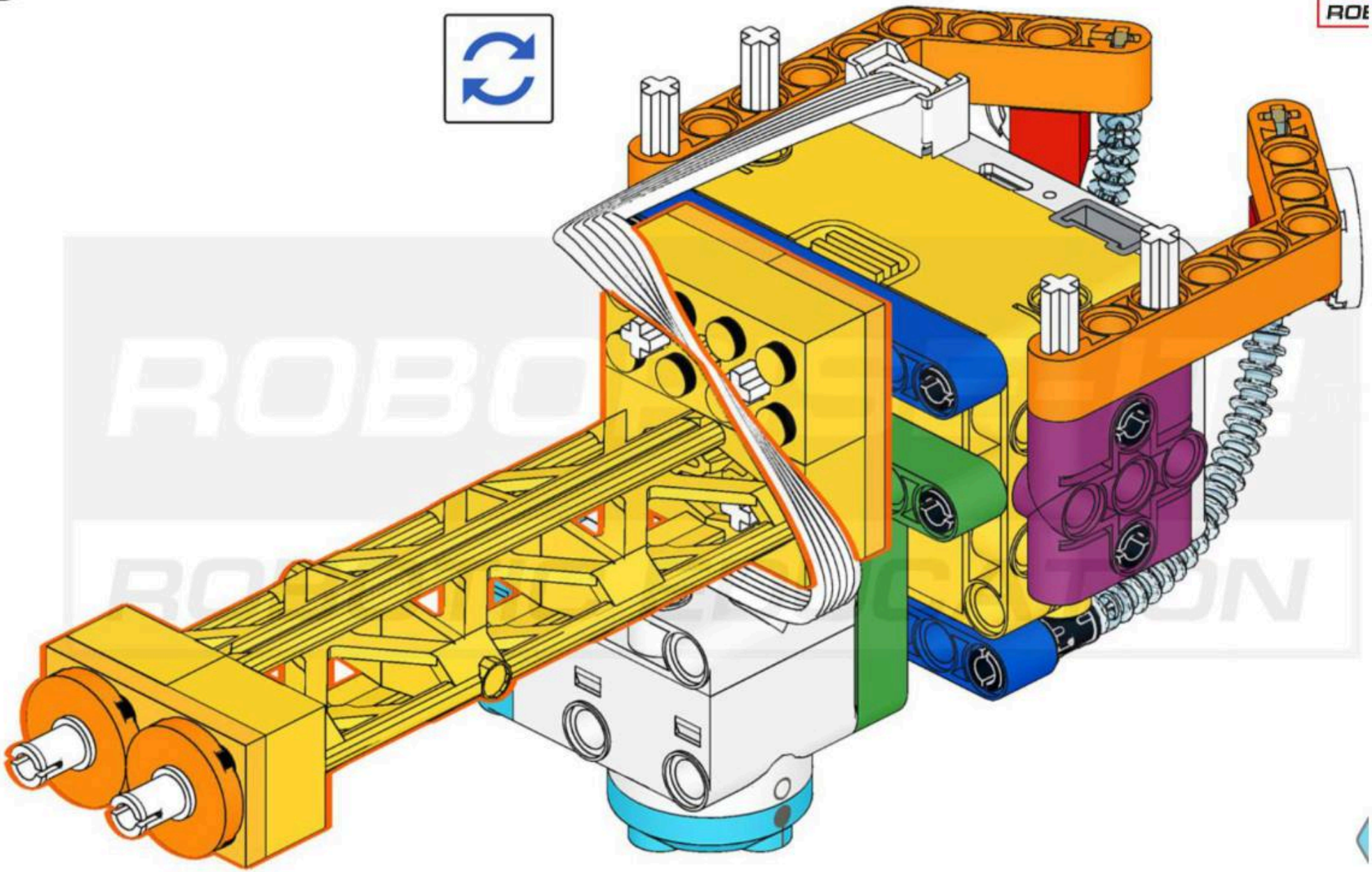


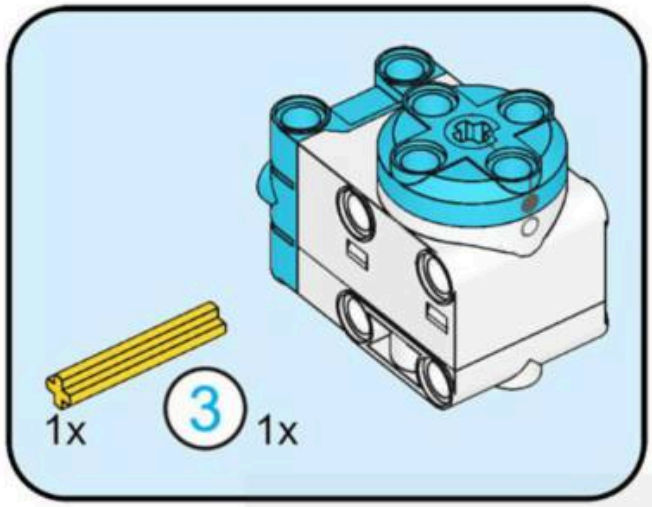
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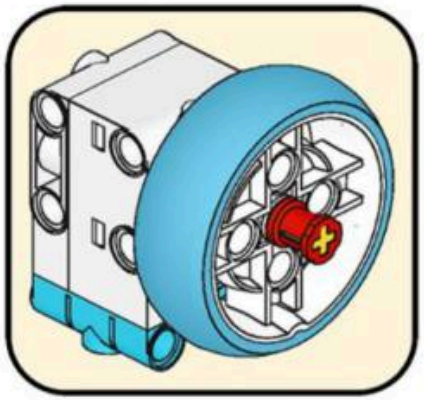


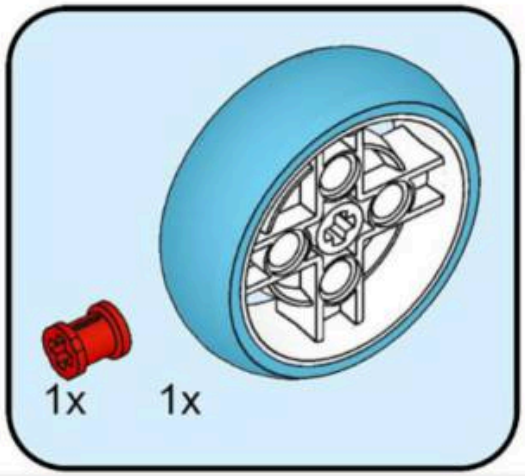


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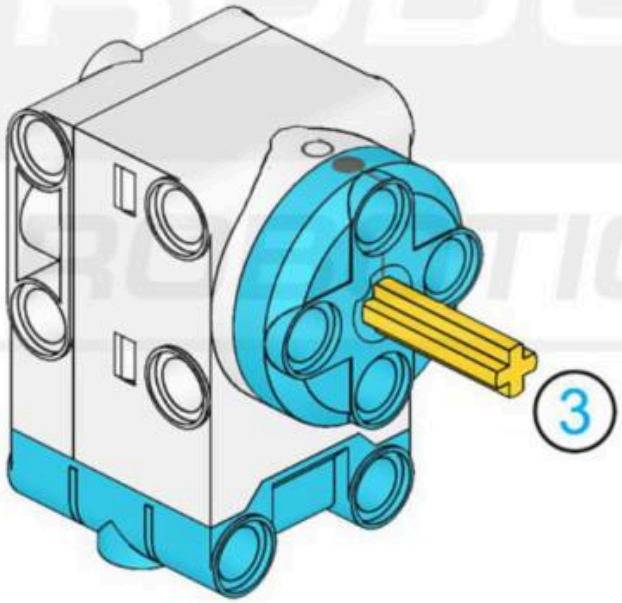




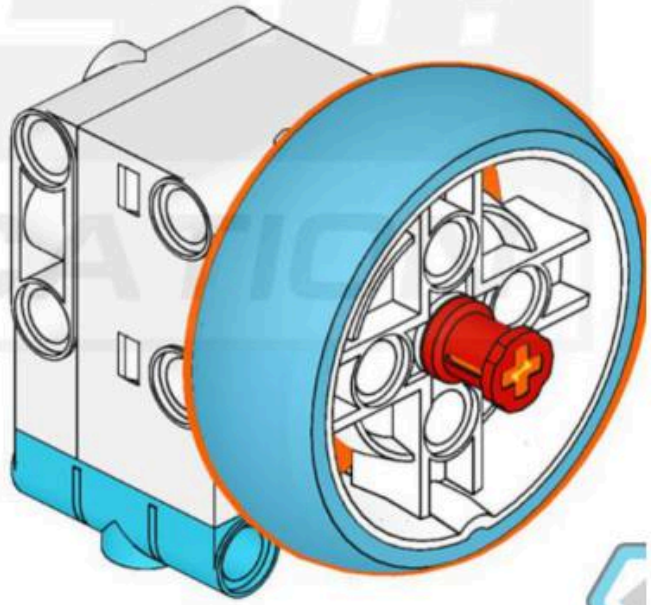




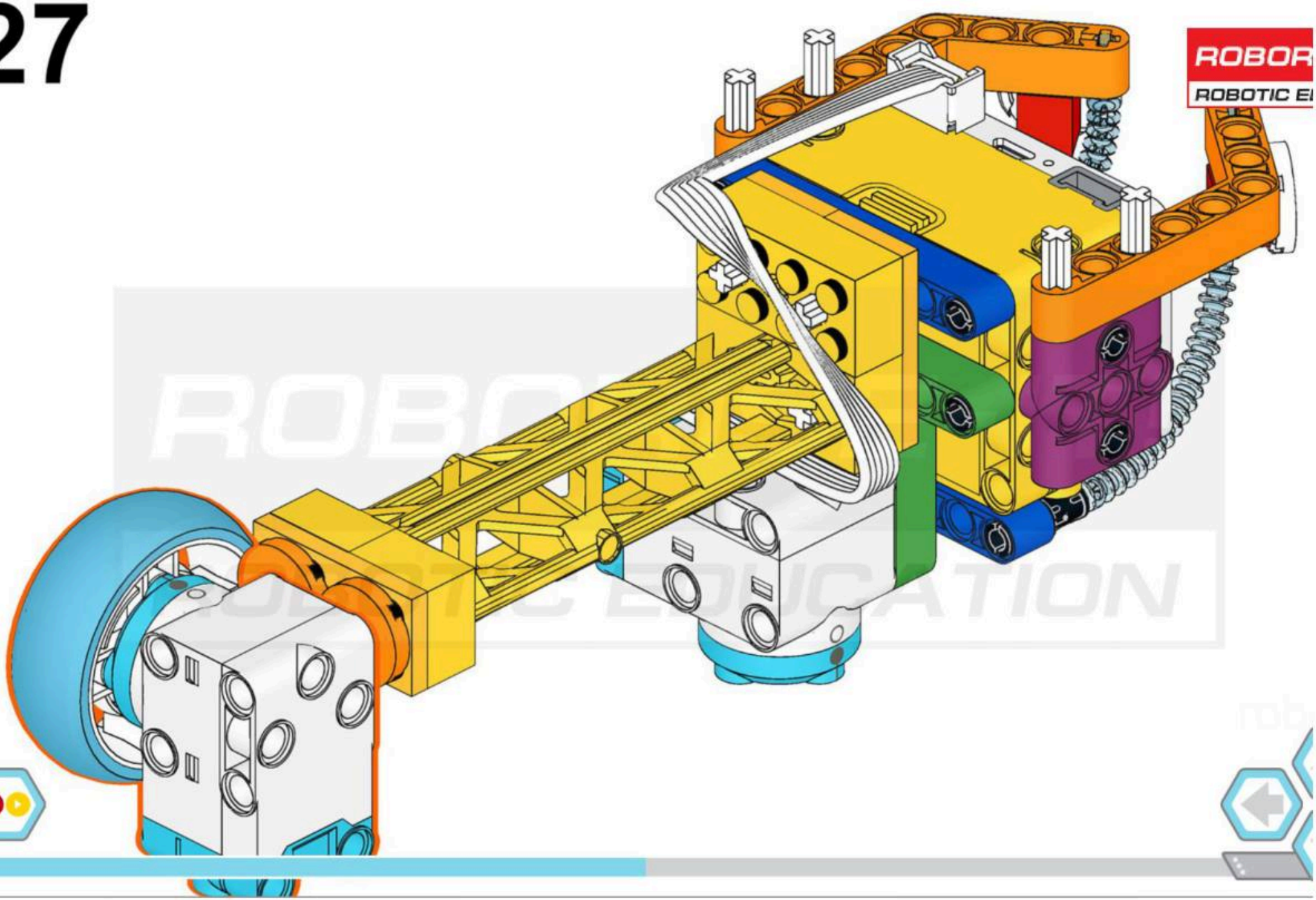
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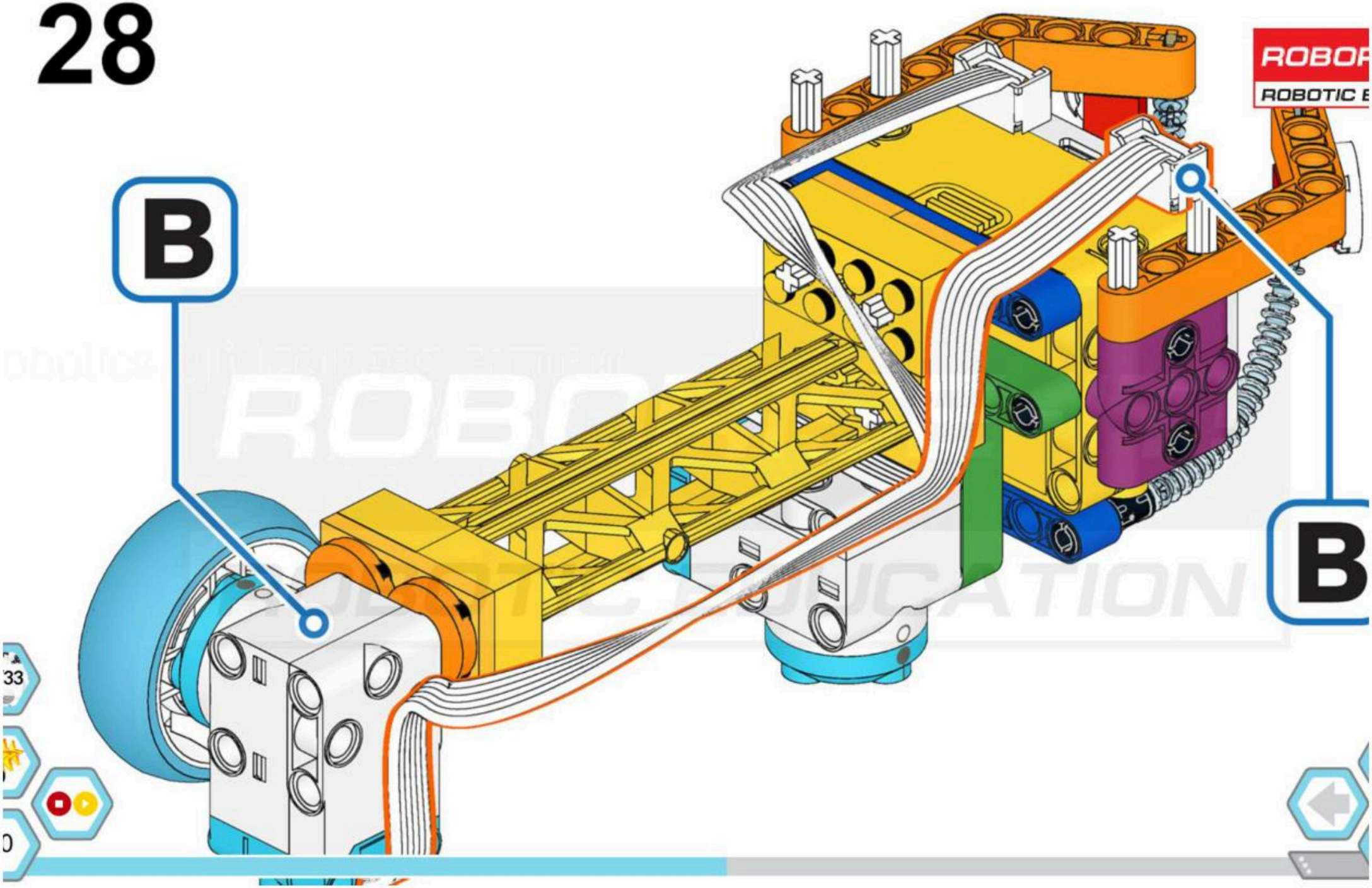


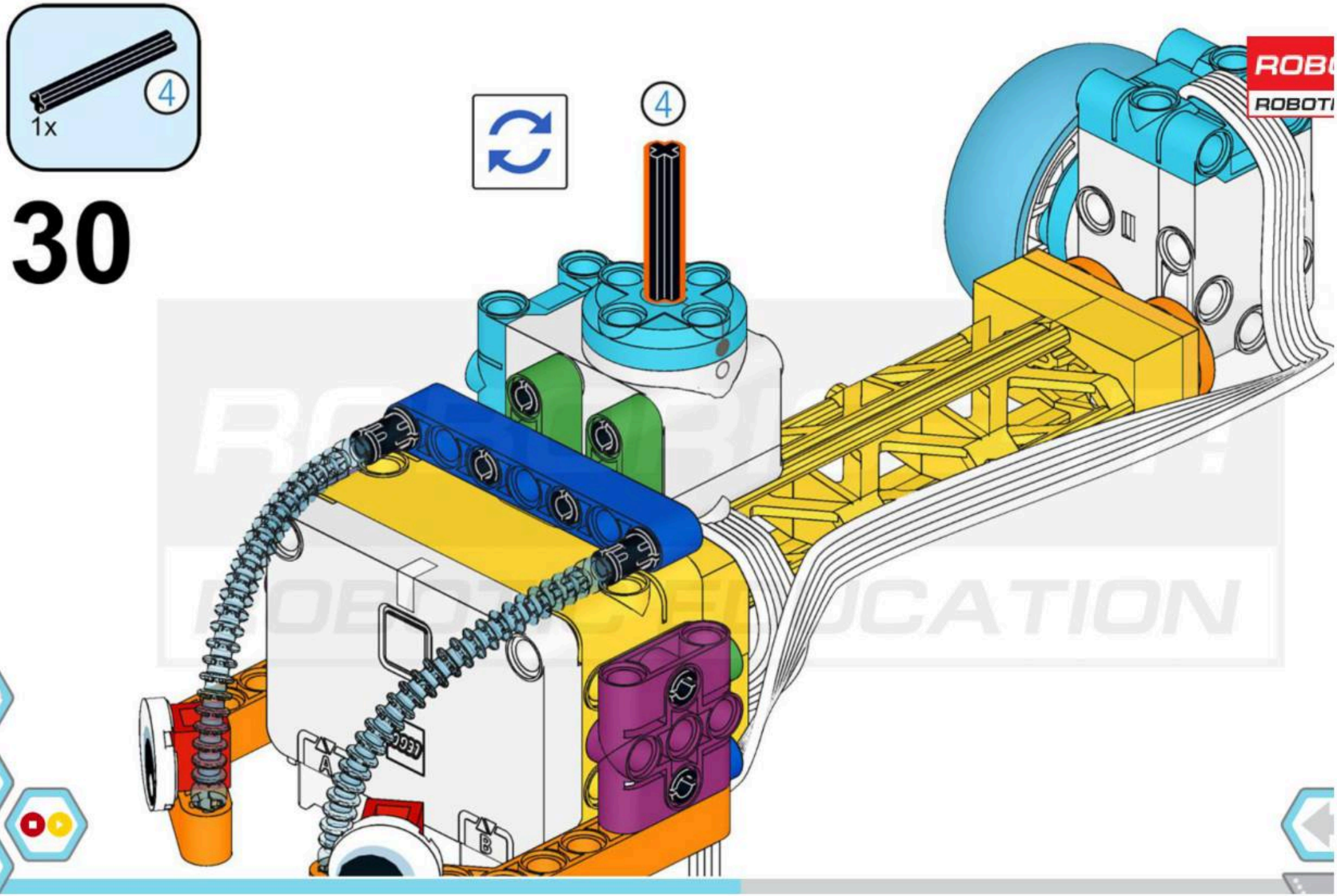
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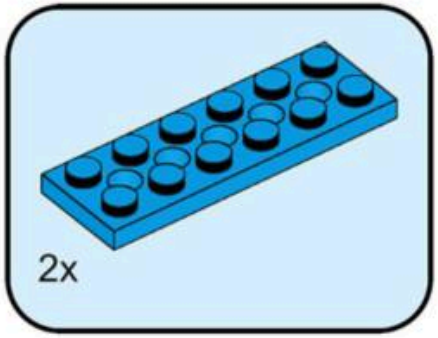


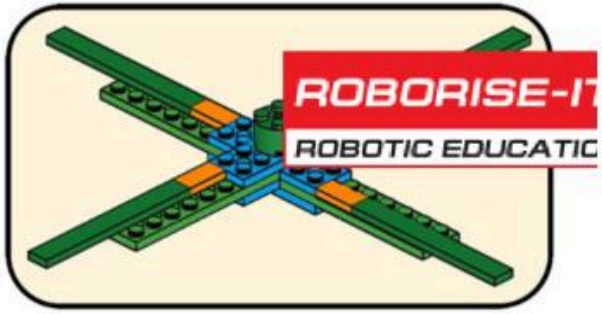
ROBOTIC EDUCATION

28

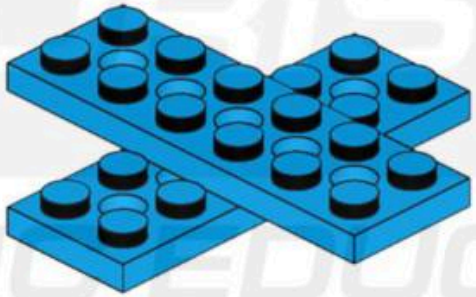



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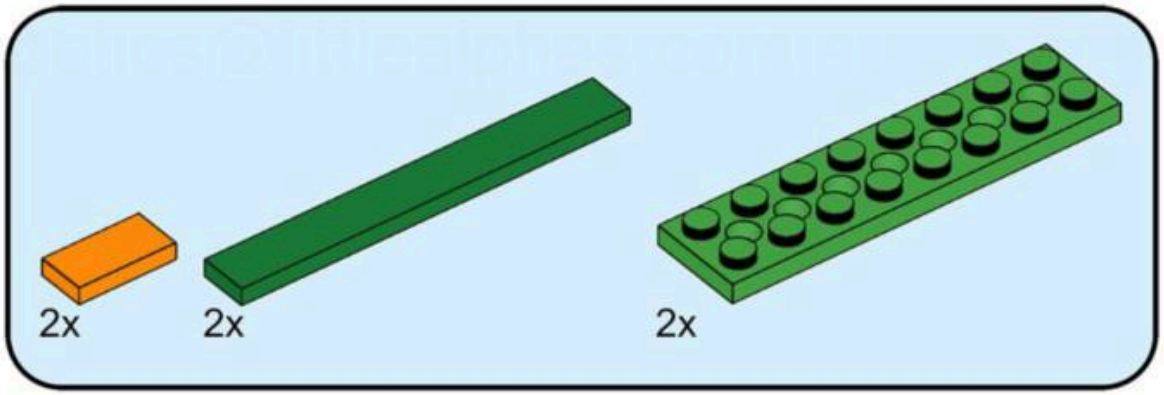




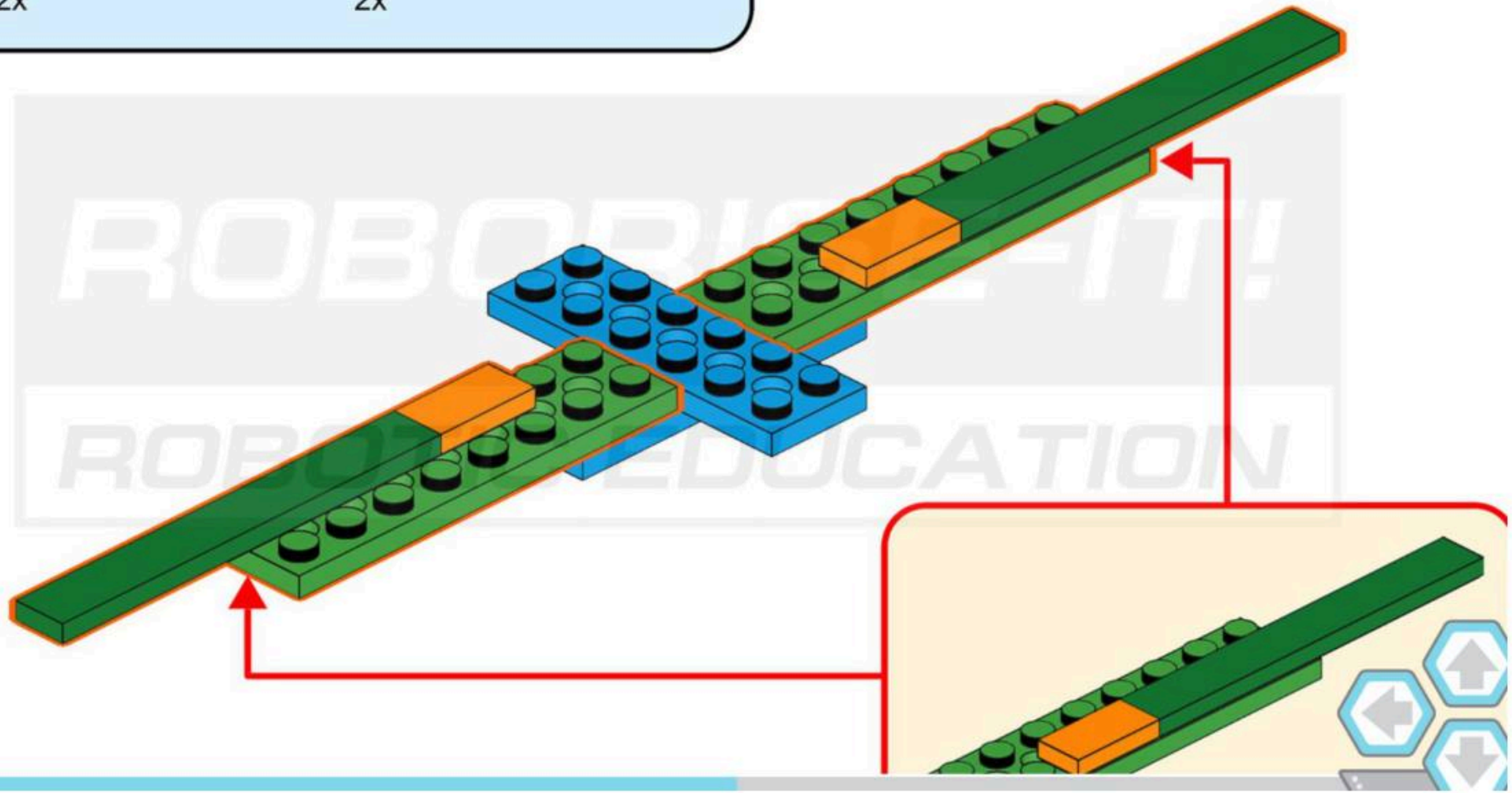
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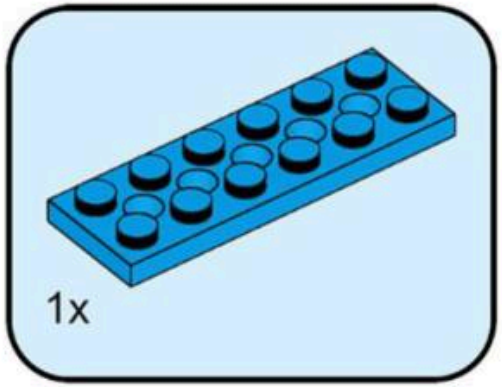
ROBORISE-IT!
ROBOTIC EDUCATION



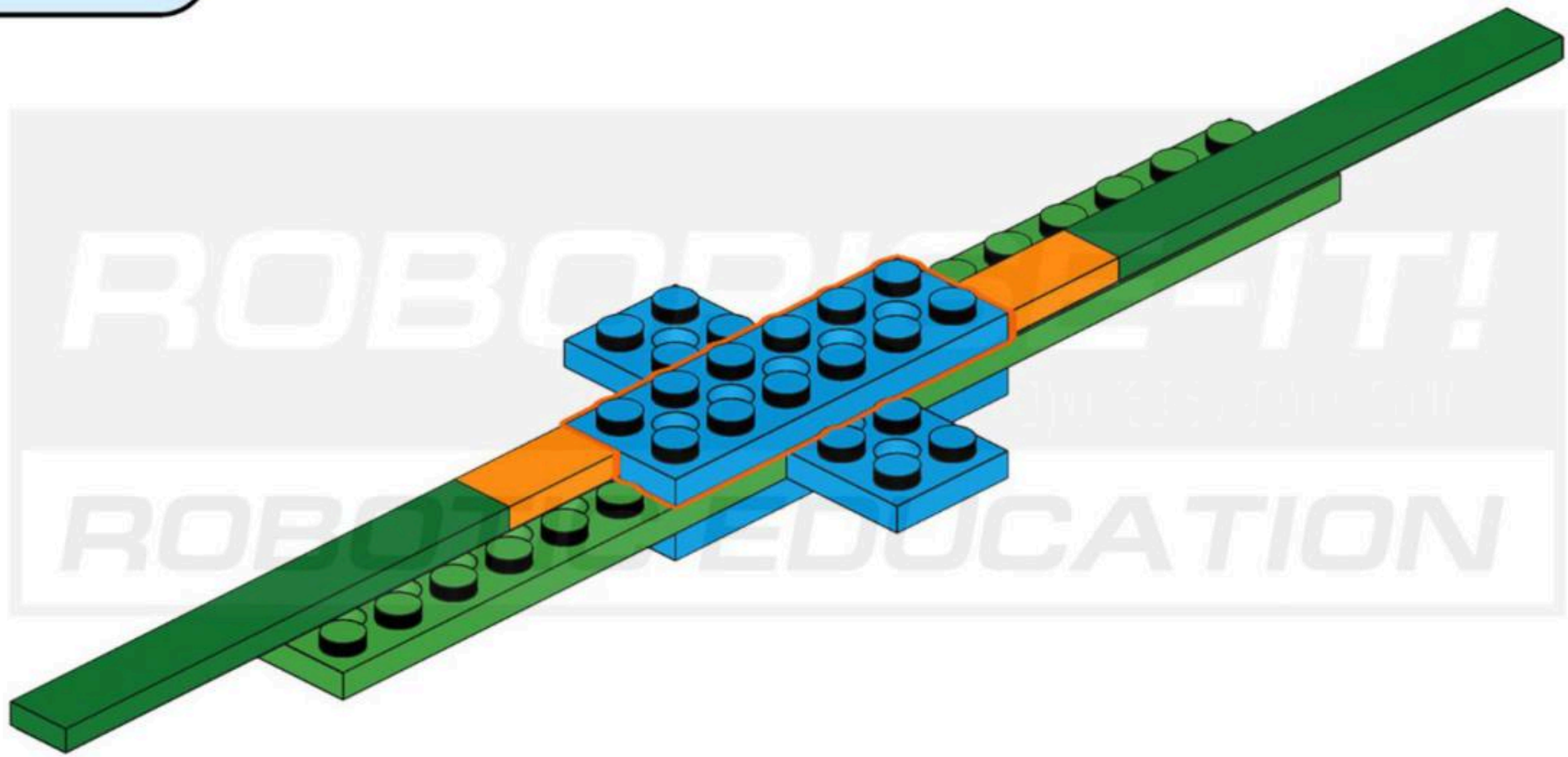
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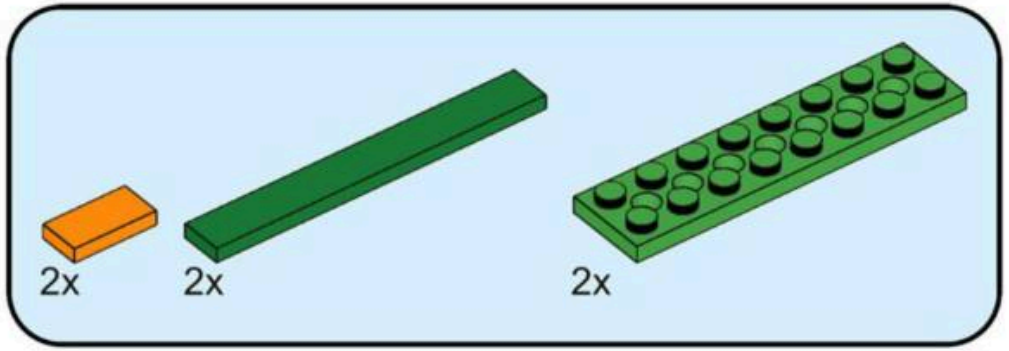
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ROBOTIC EDUCATION



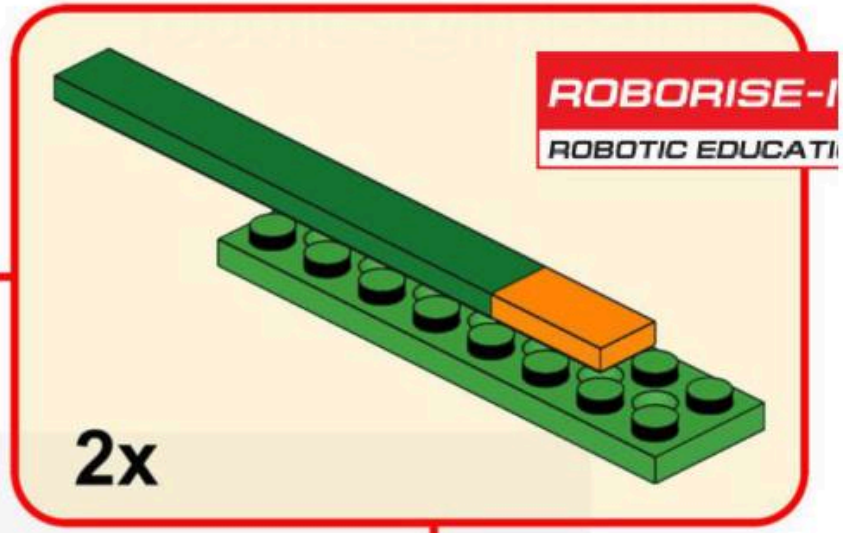
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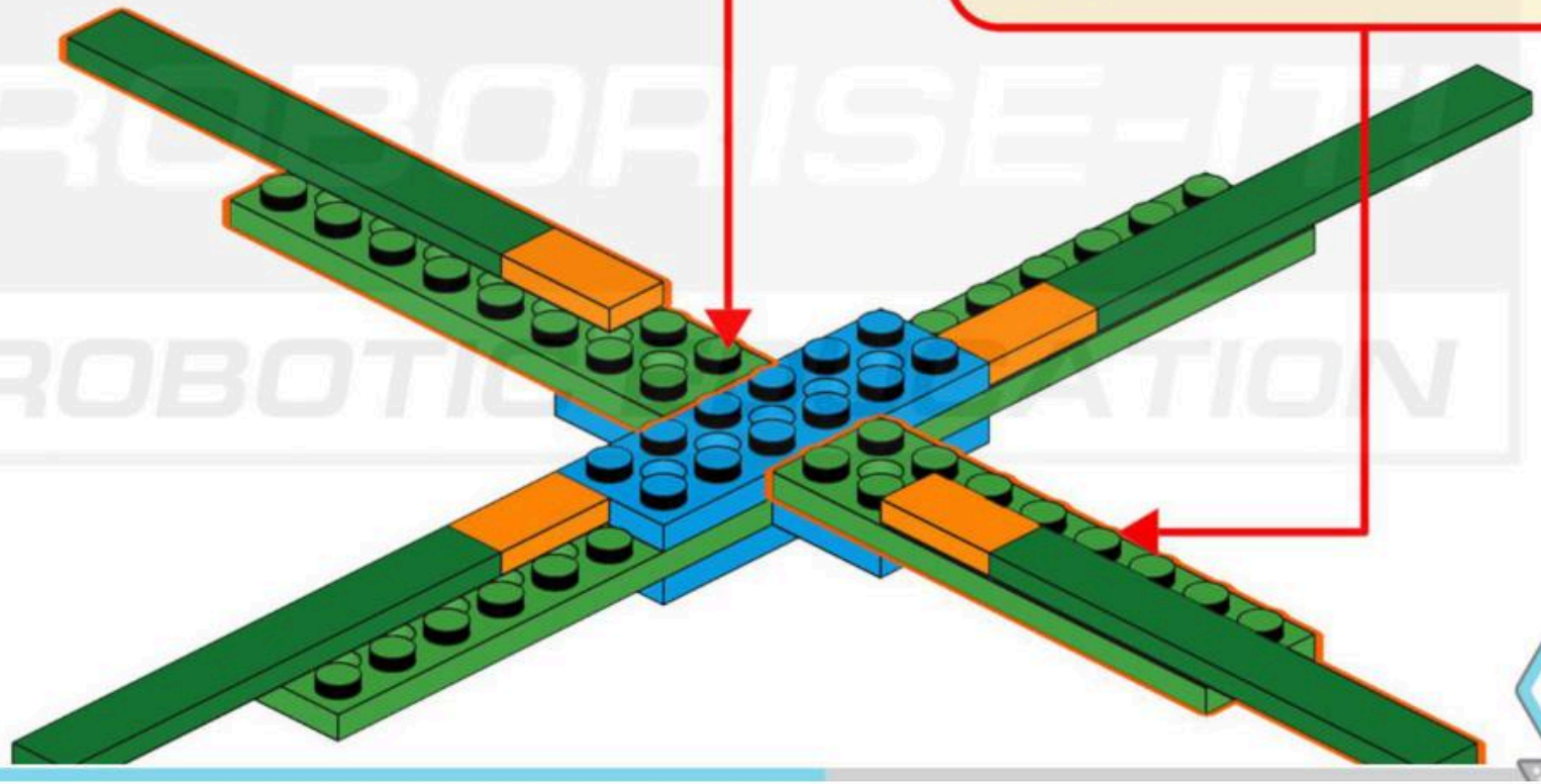


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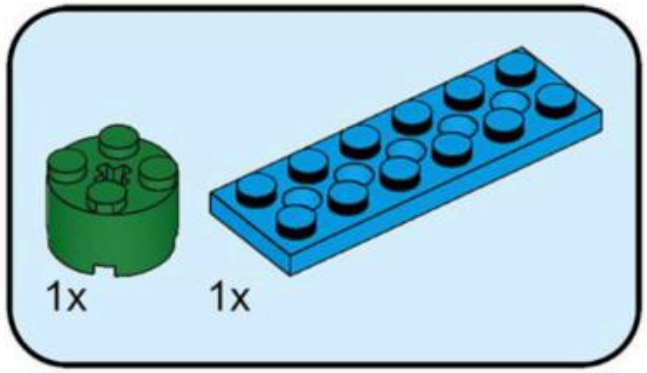


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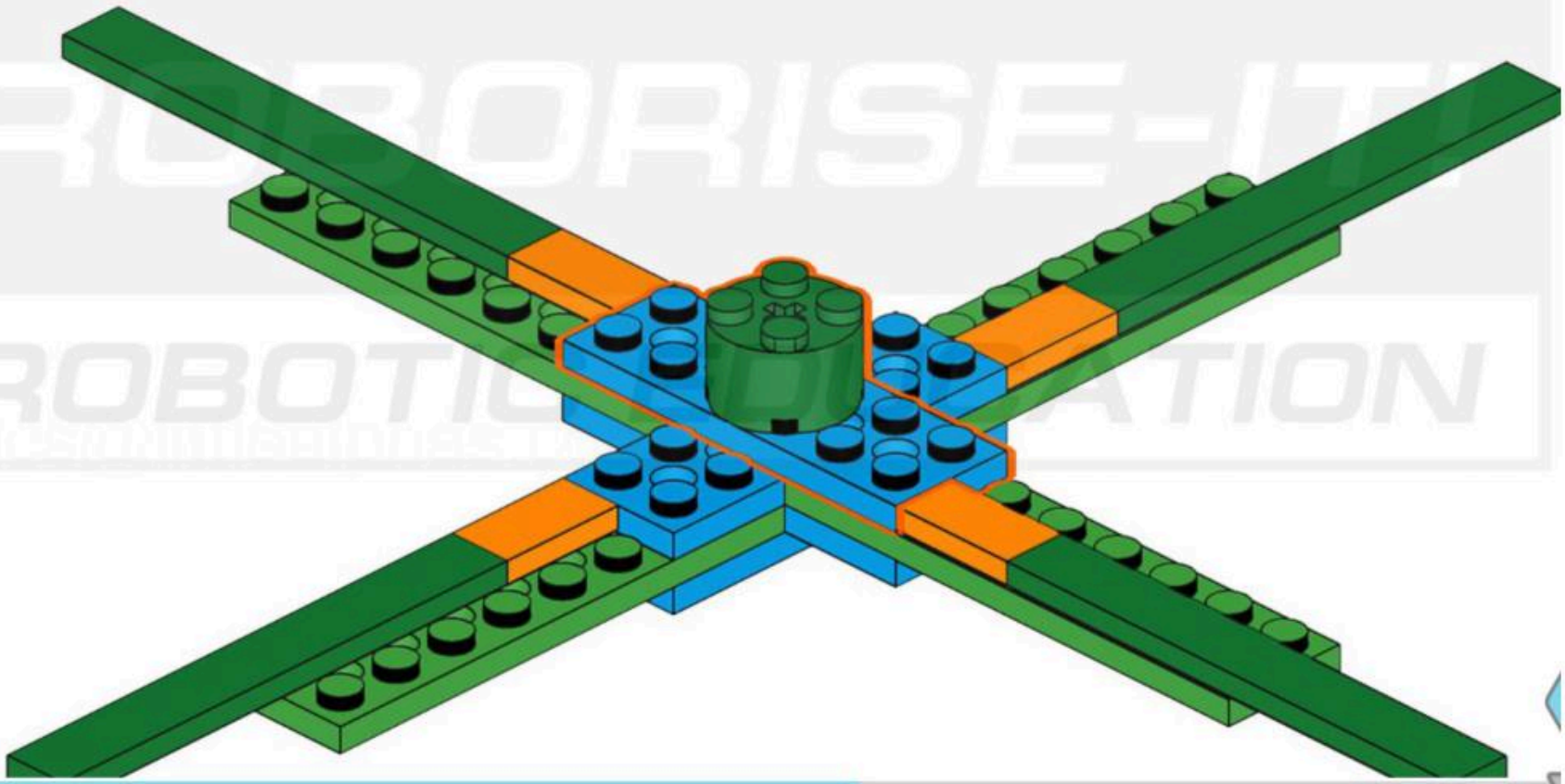




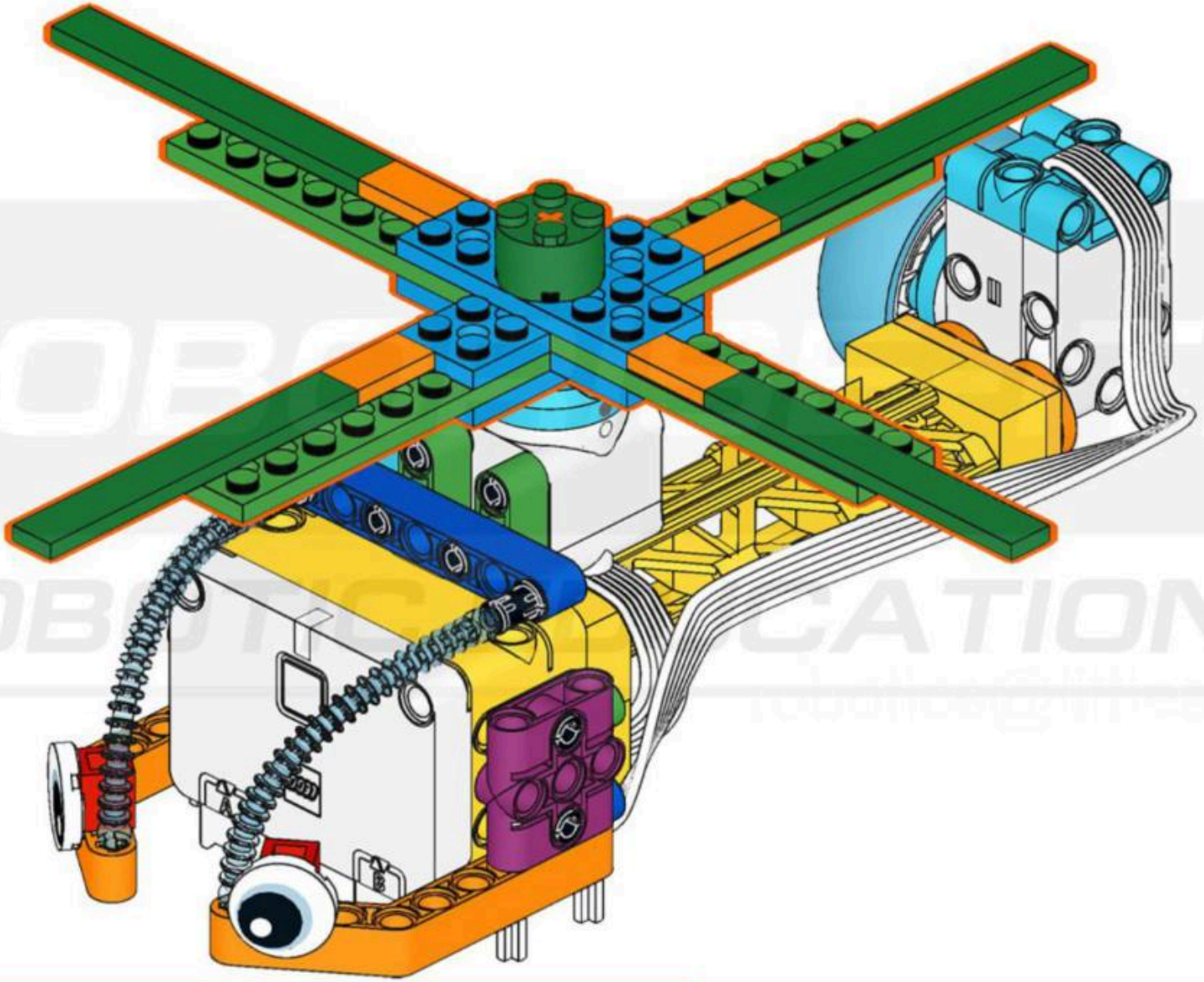
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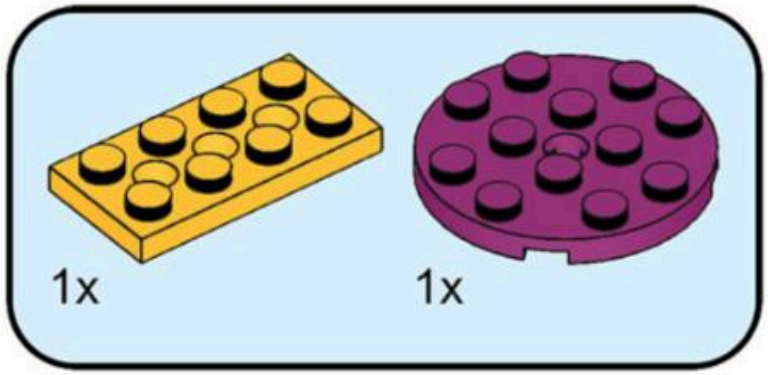


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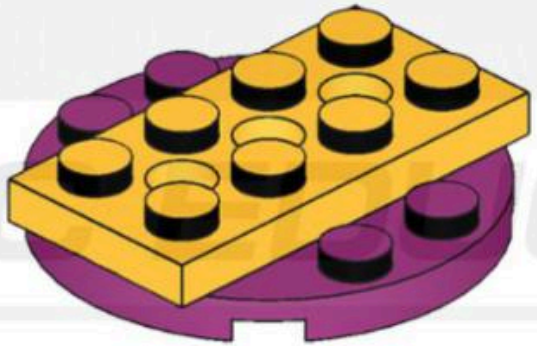
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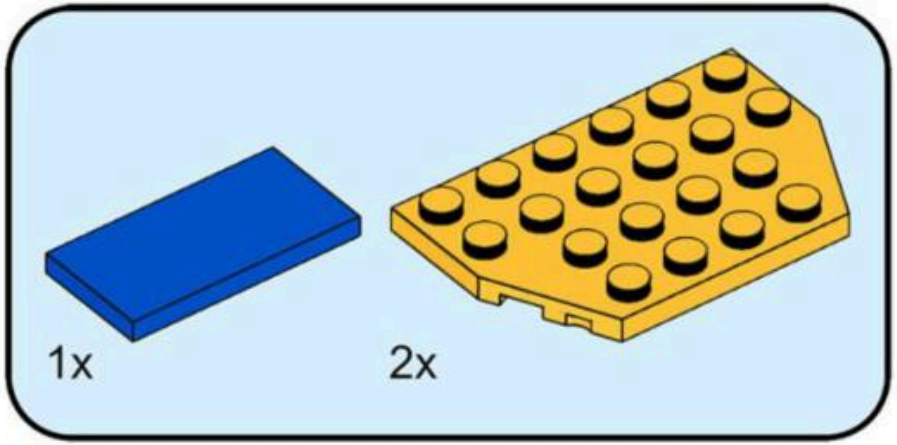
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ROBORISE
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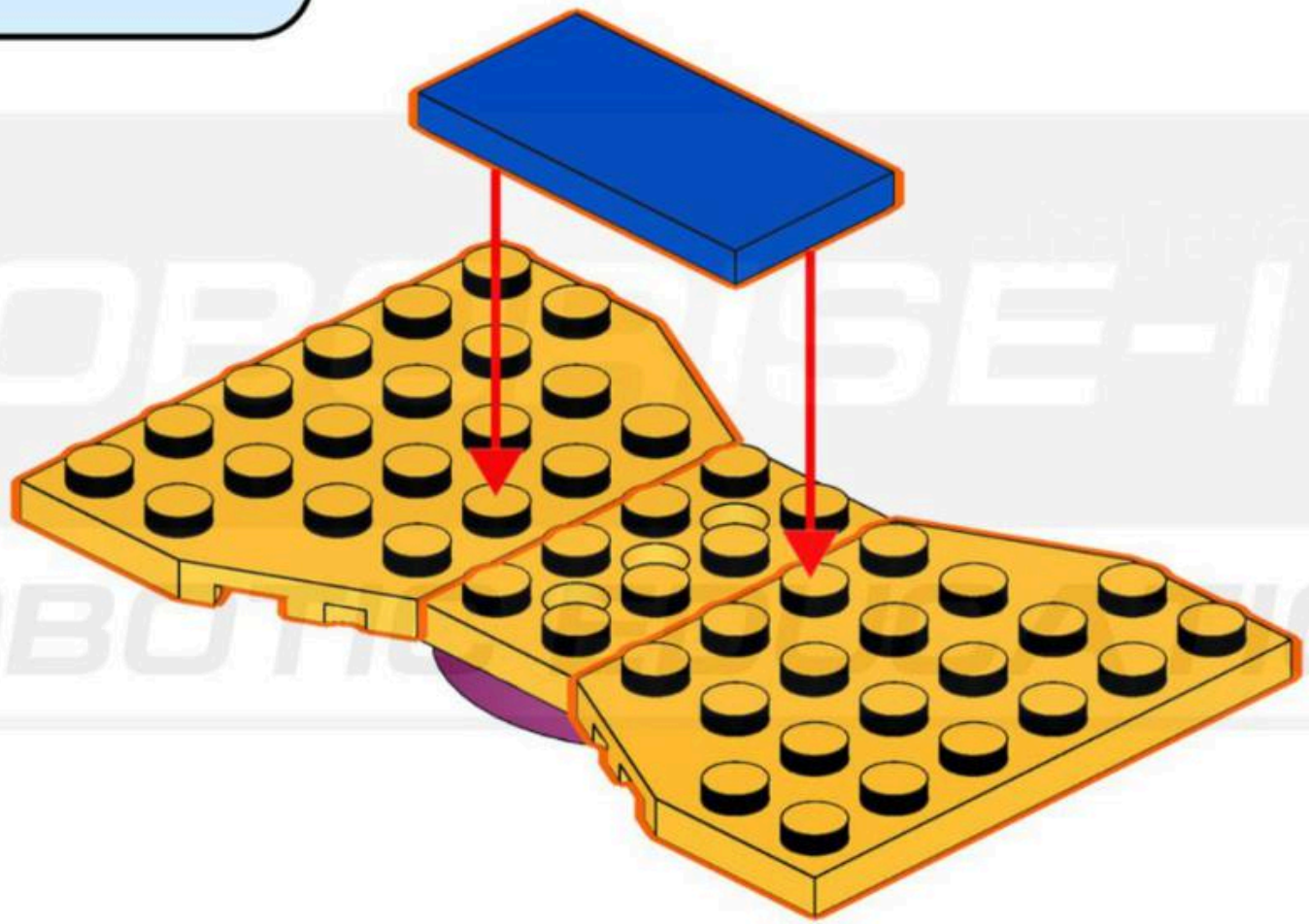


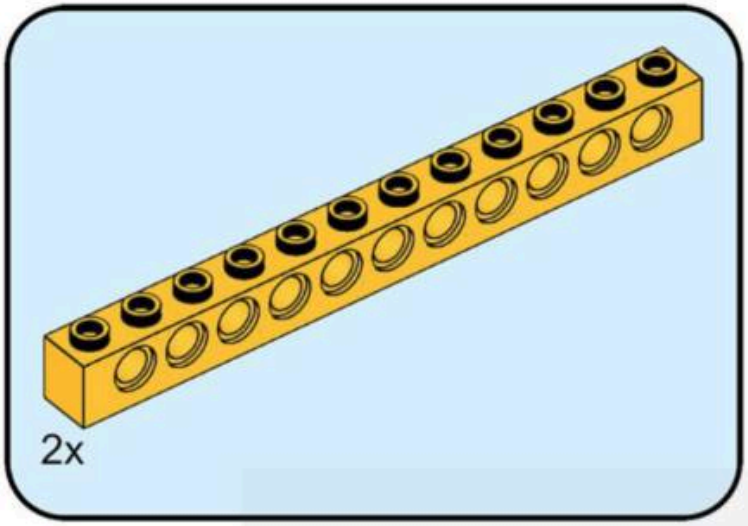




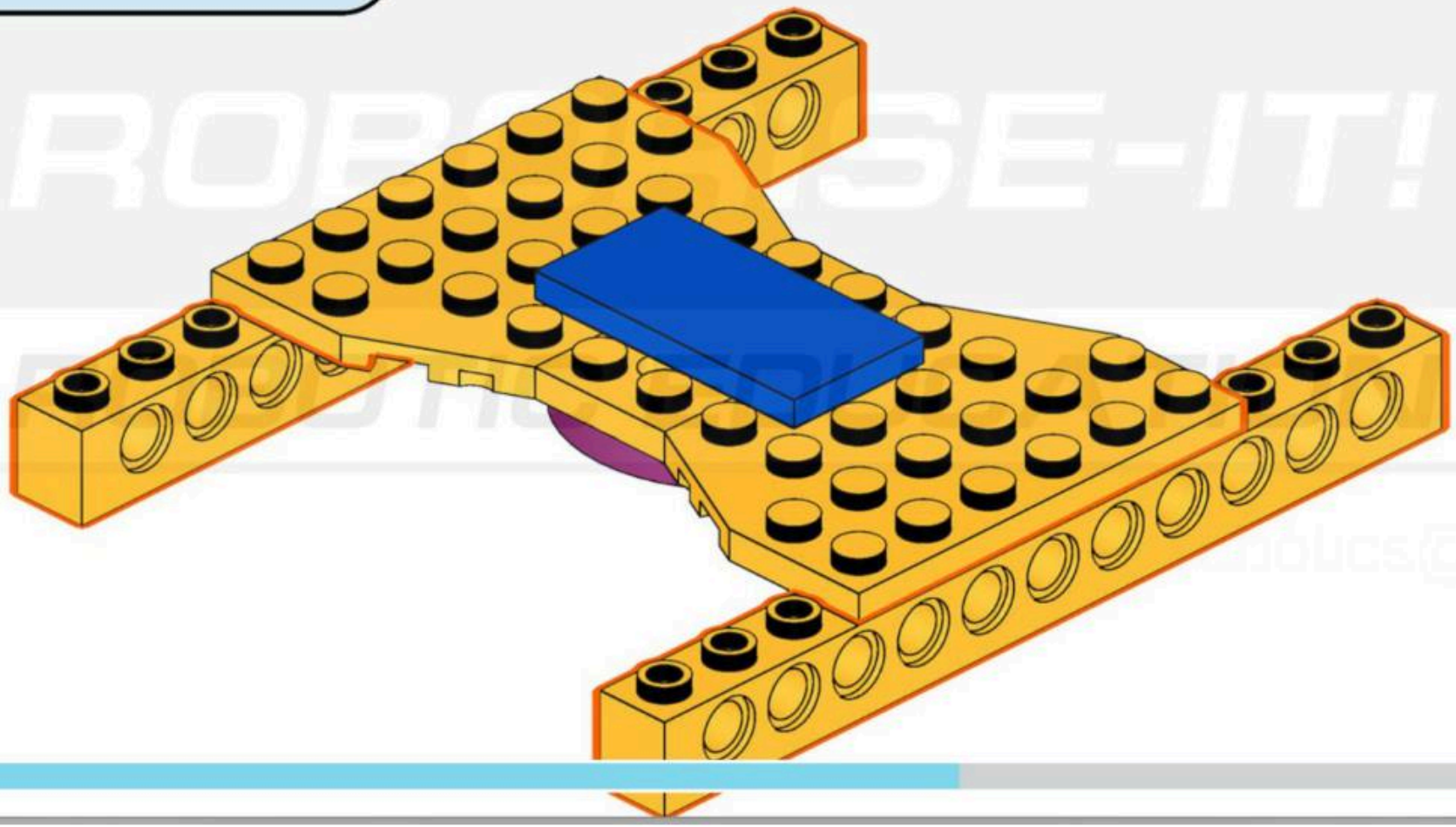


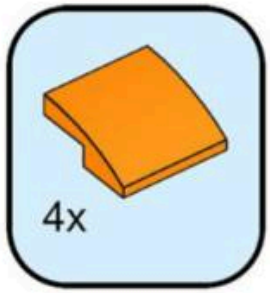
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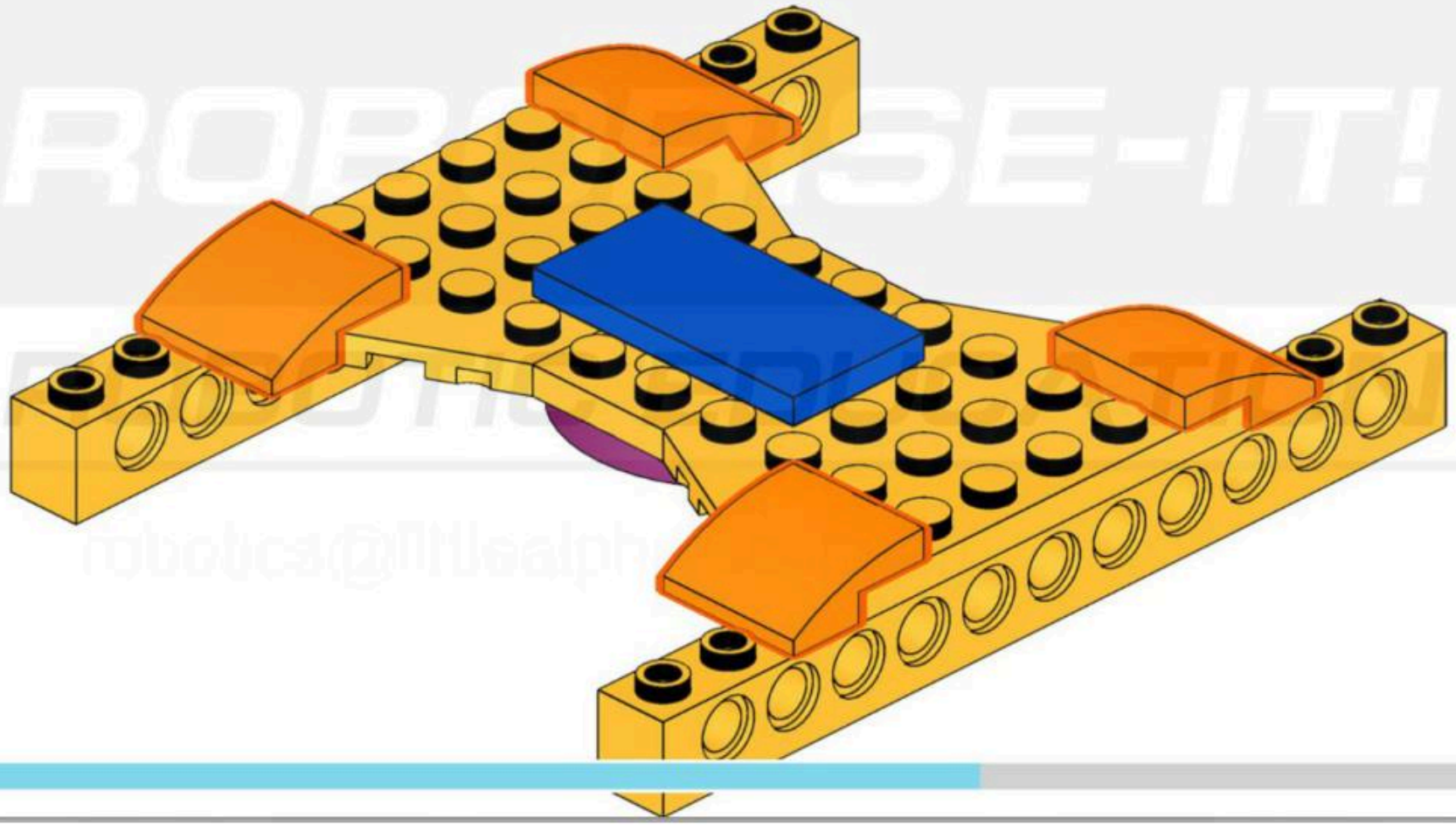


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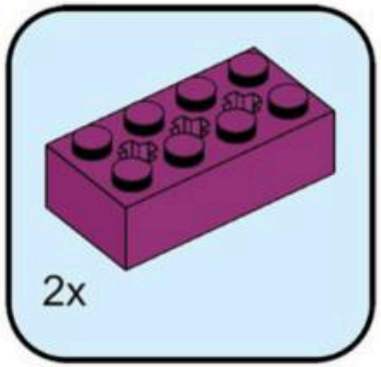




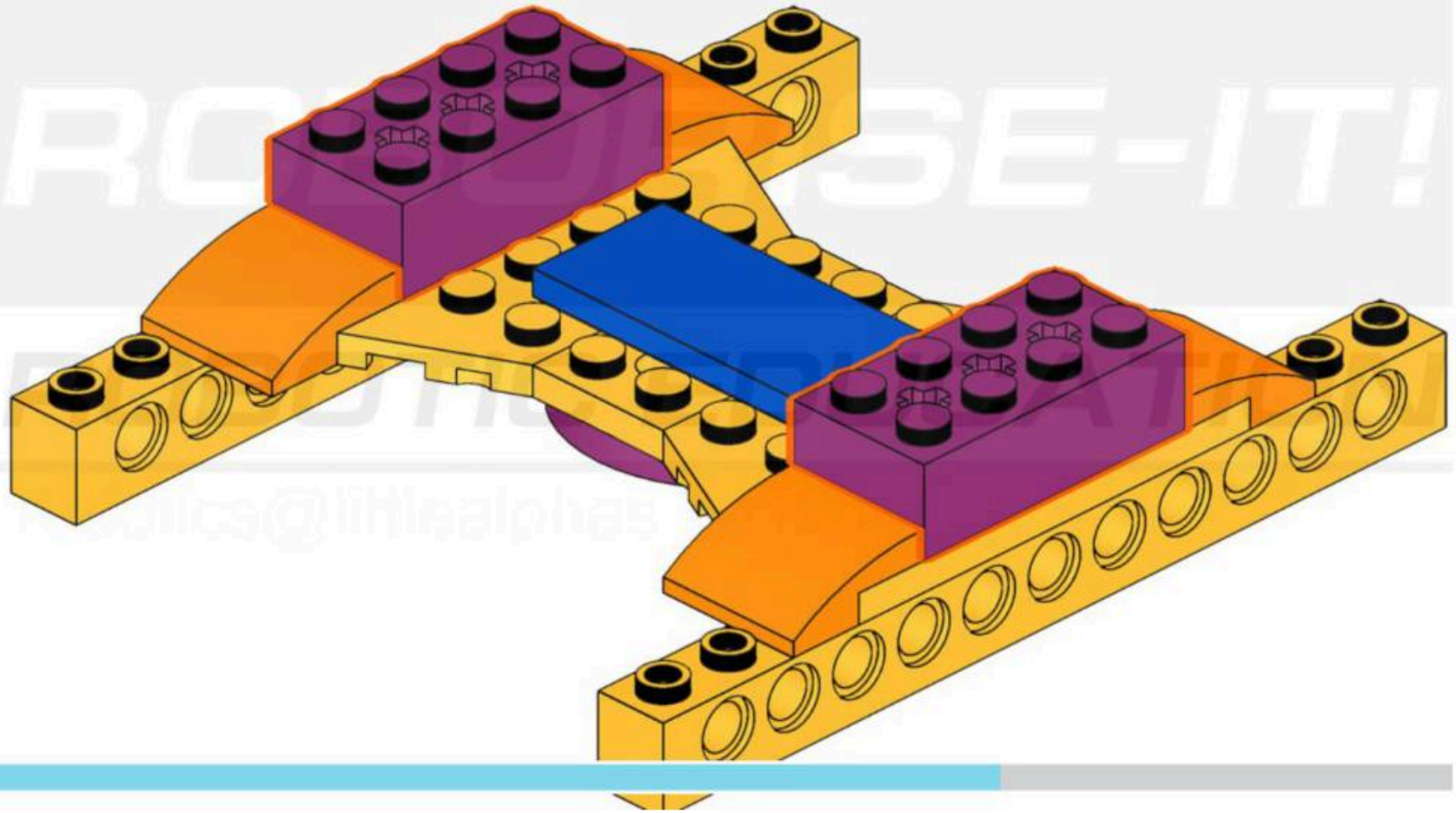
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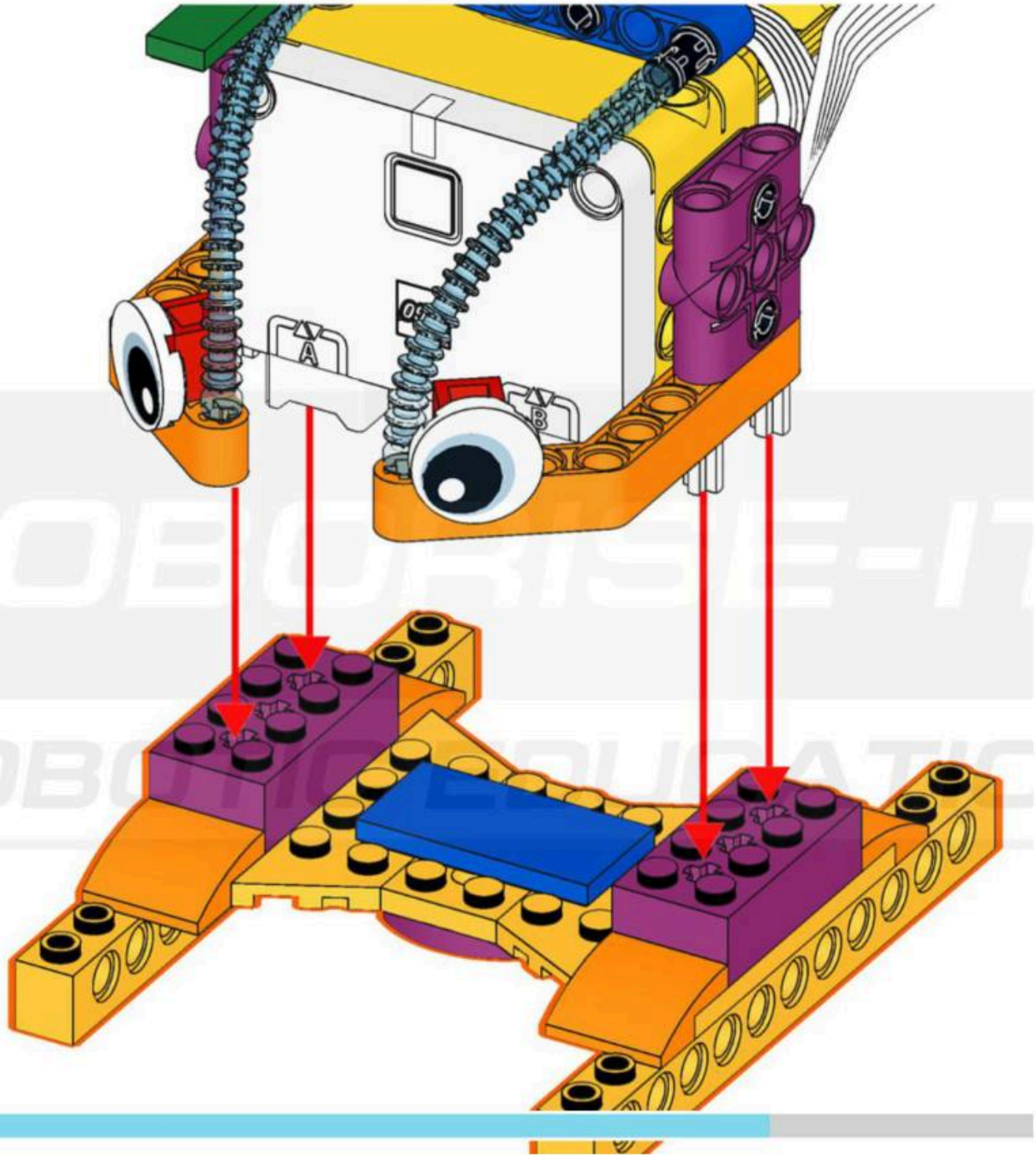
ROBOTICS-IT!



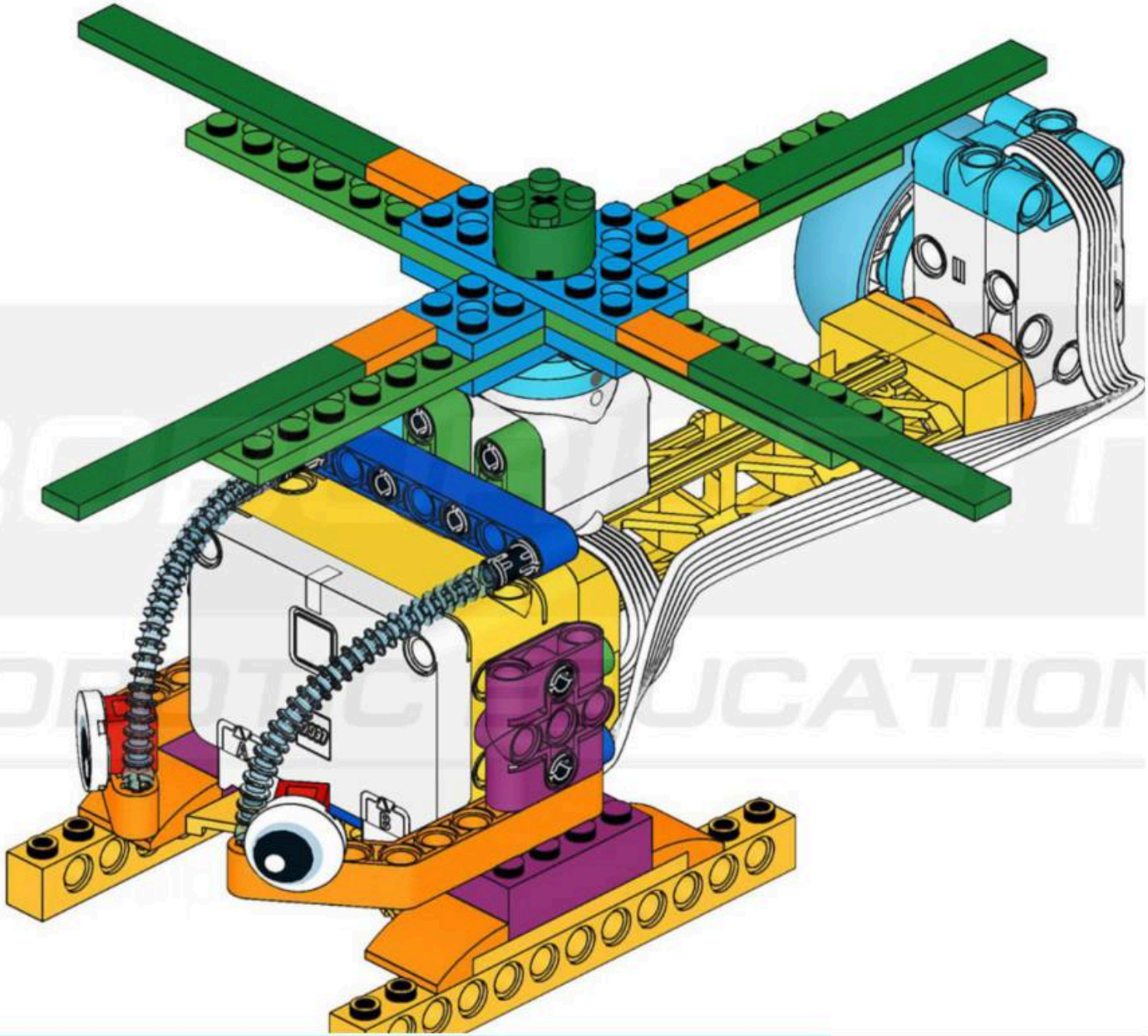
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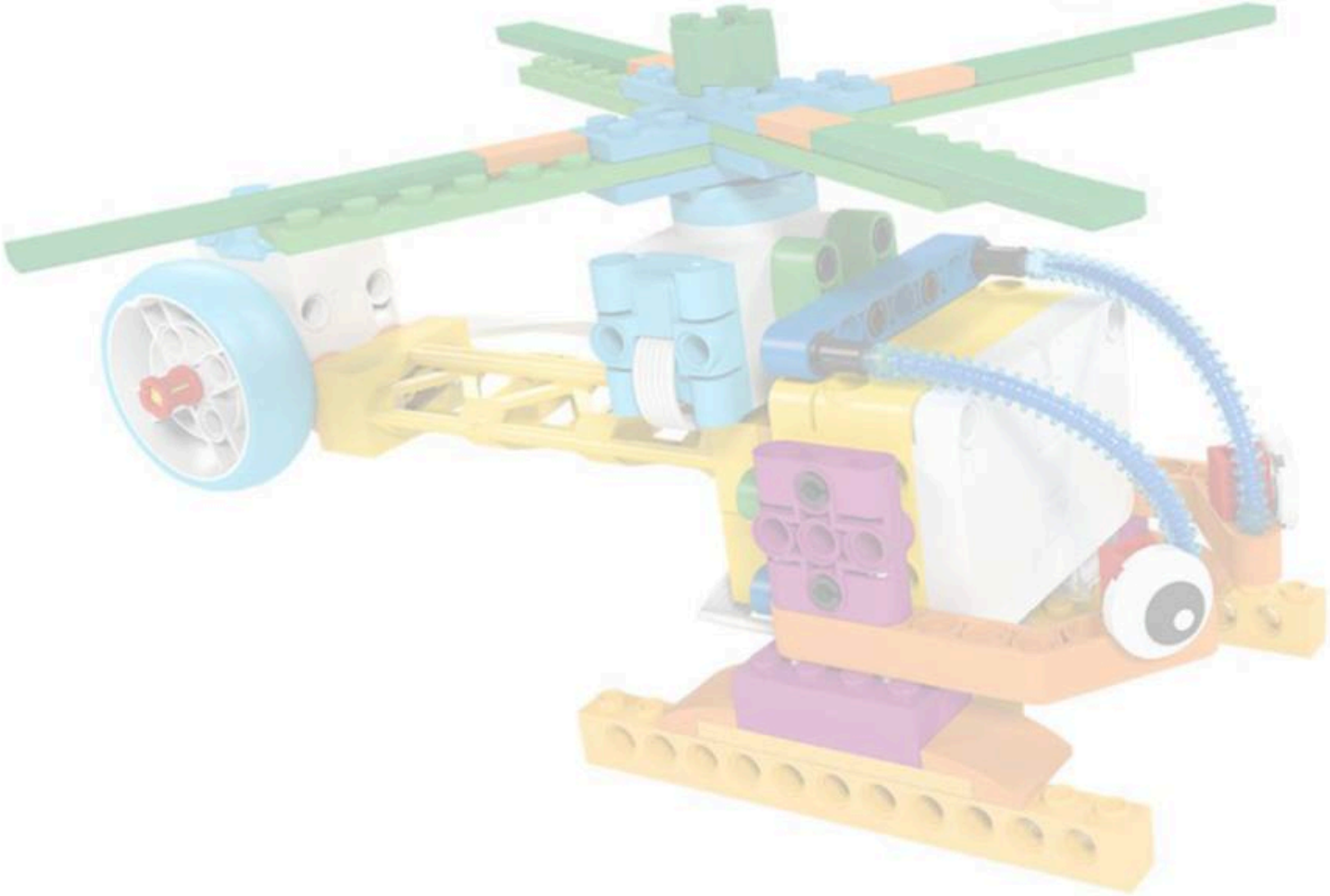
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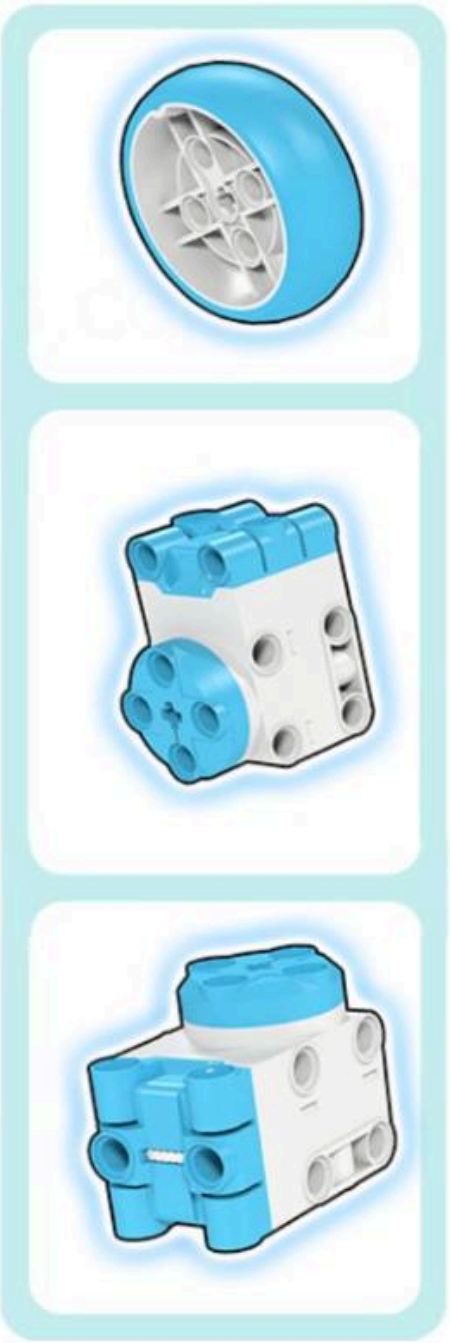


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Place the parts of the robot in the correct places







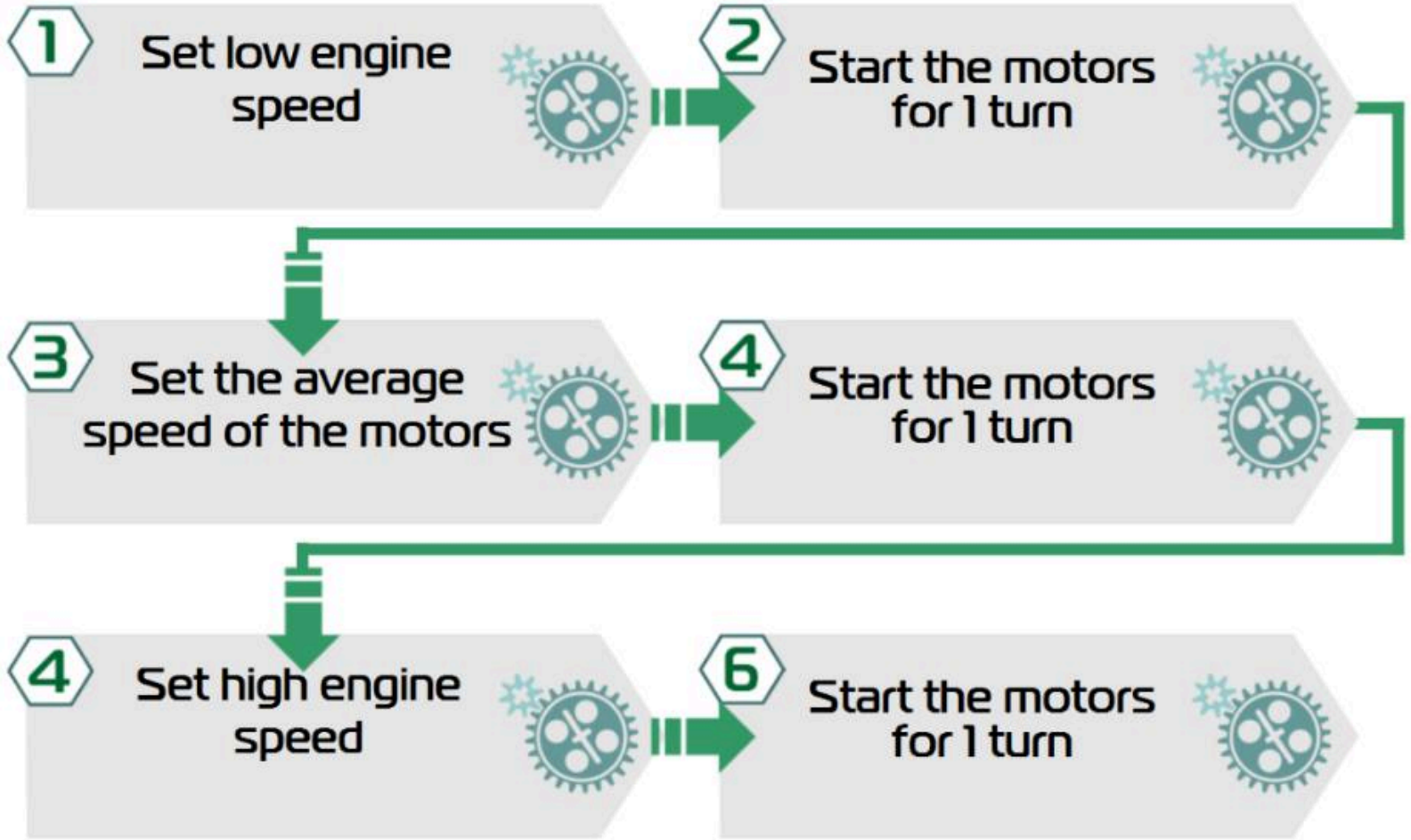
Remember what sensors the Robot Hub contains. What are they called and how do they differ from each other

With a sudden change in rotation speed, all mechanisms are under increased load, which is not desirable. Program a gradual increase in the rotation speed of the robot's screws



Task 1. Algorithm

The simplest algorithm can look like this:



Task 1. Program

ICON BLOCKS | WORD BLOCKS

Use the motor blocks to program the robot's movement:

In ICON Blocks mode, the program looks like this:

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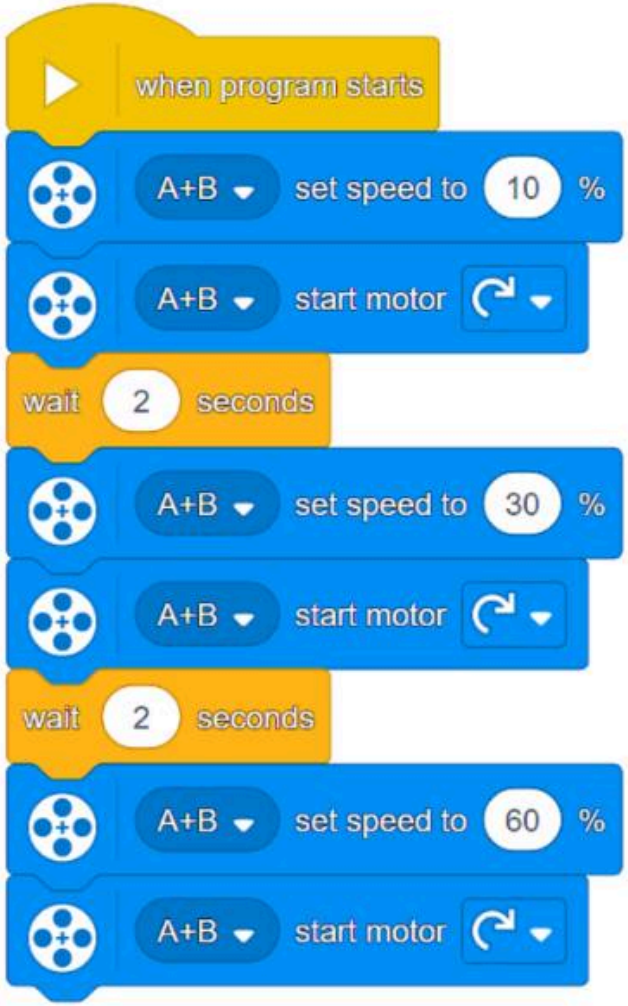


Task 1. Program

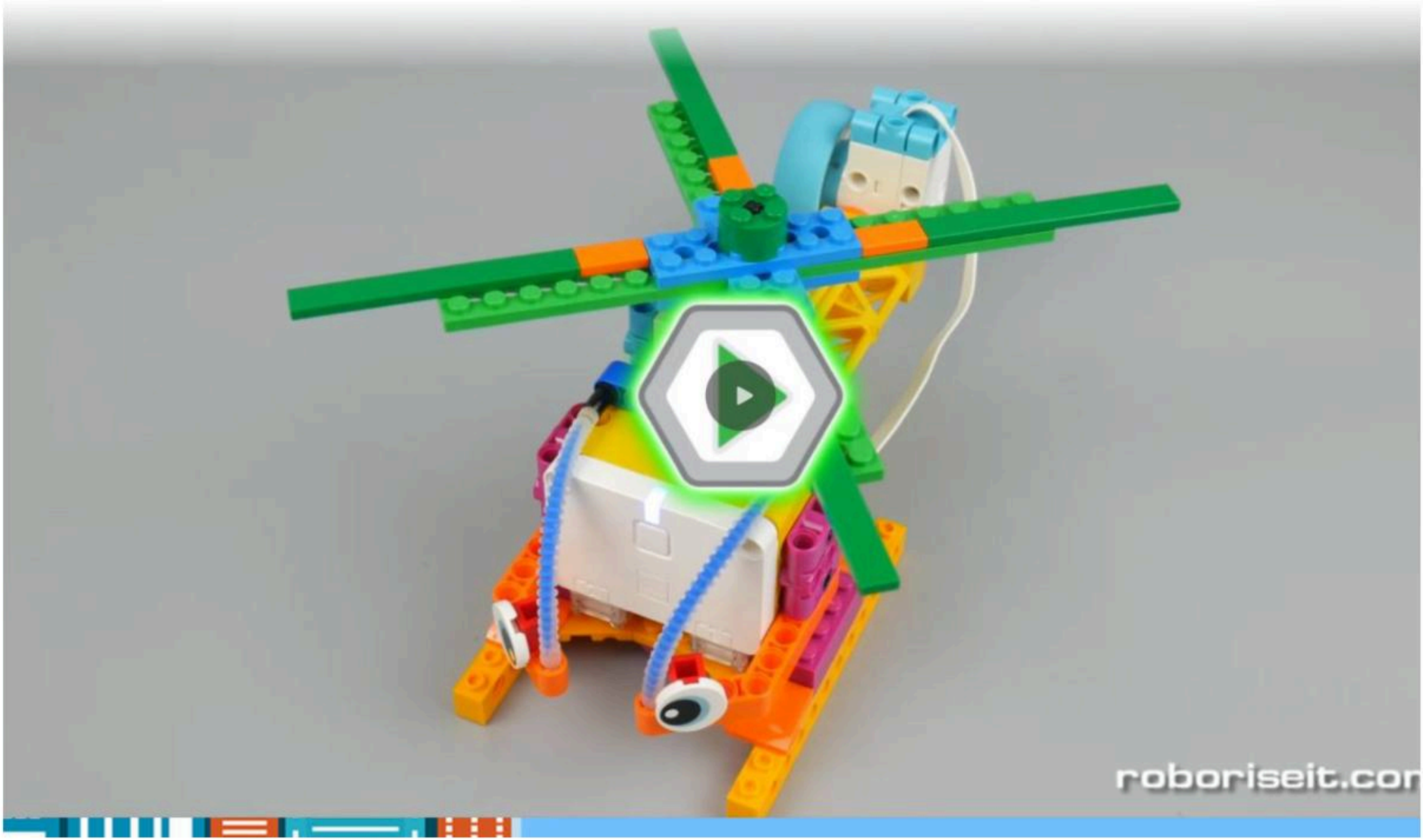
ICON BLOCKS | WORD BLOCKS

Use the motor blocks to program the robot's movement:

In Word Blocks mode, the program looks like this:



When testing in ICON Blocks mode, you may have noticed that instead of a smooth acceleration, there were three separate stages of rotation with stops in between. Let's get rid of those stops and make acceleration smooth.



ICON BLOCKS WORD BLOCKS

Create three programs that will run simultaneously.

In ICON Blocks mode, the program looks like this:







So the robot will start with the lowest rotation speed, after a second it will increase it, and after another second there will be a second increase in rotation speed

Due to the presence of an accelerometer, we can use its data to change the speed of rotation of the propellers when changing the position of the helicopter. Program the corresponding speed for each position of the robot



Use the motor blocks to program the rotation of the screws:

In ICON Blocks mode, programs look like this:

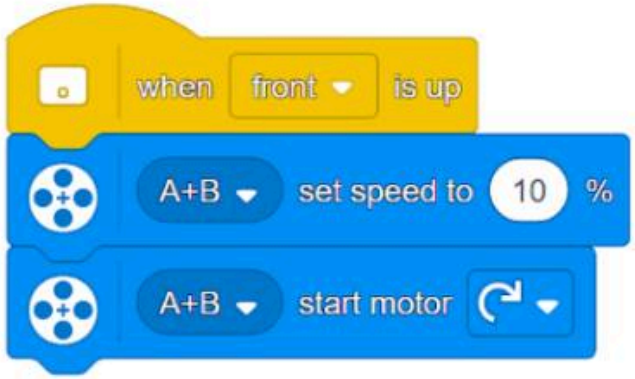


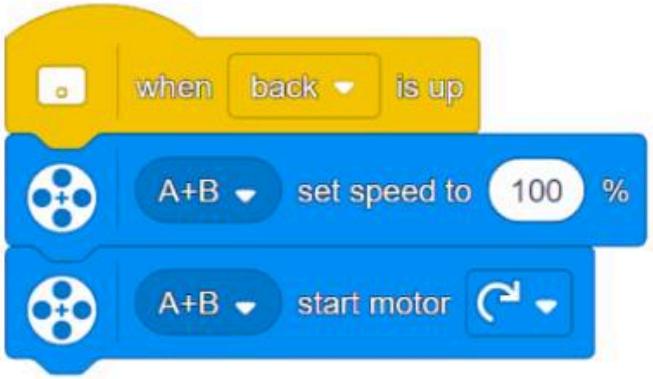


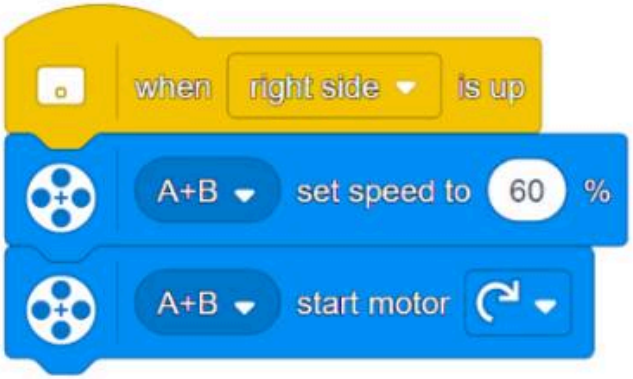


Use the motor blocks to program the rotation of the screws:

In Word Blocks mode, programs look like this:











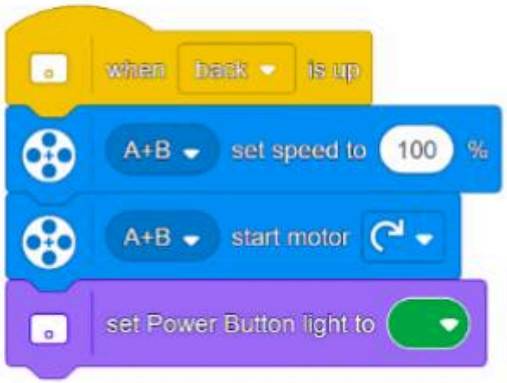
To the rest of the programs, add processing of one more position of the robot:

In ICON Blocks mode, the program looks like this:



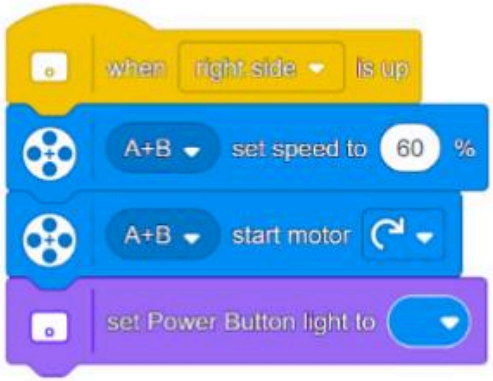
Program and test your robot.

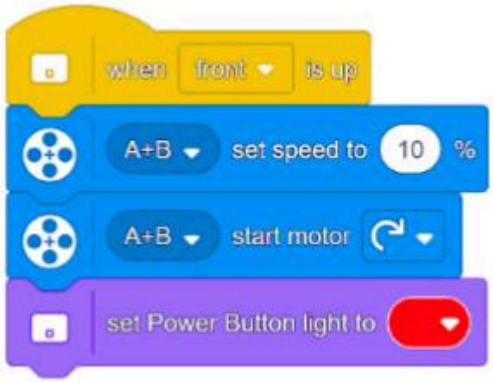
In Word Blocks mode, programs will look like this.





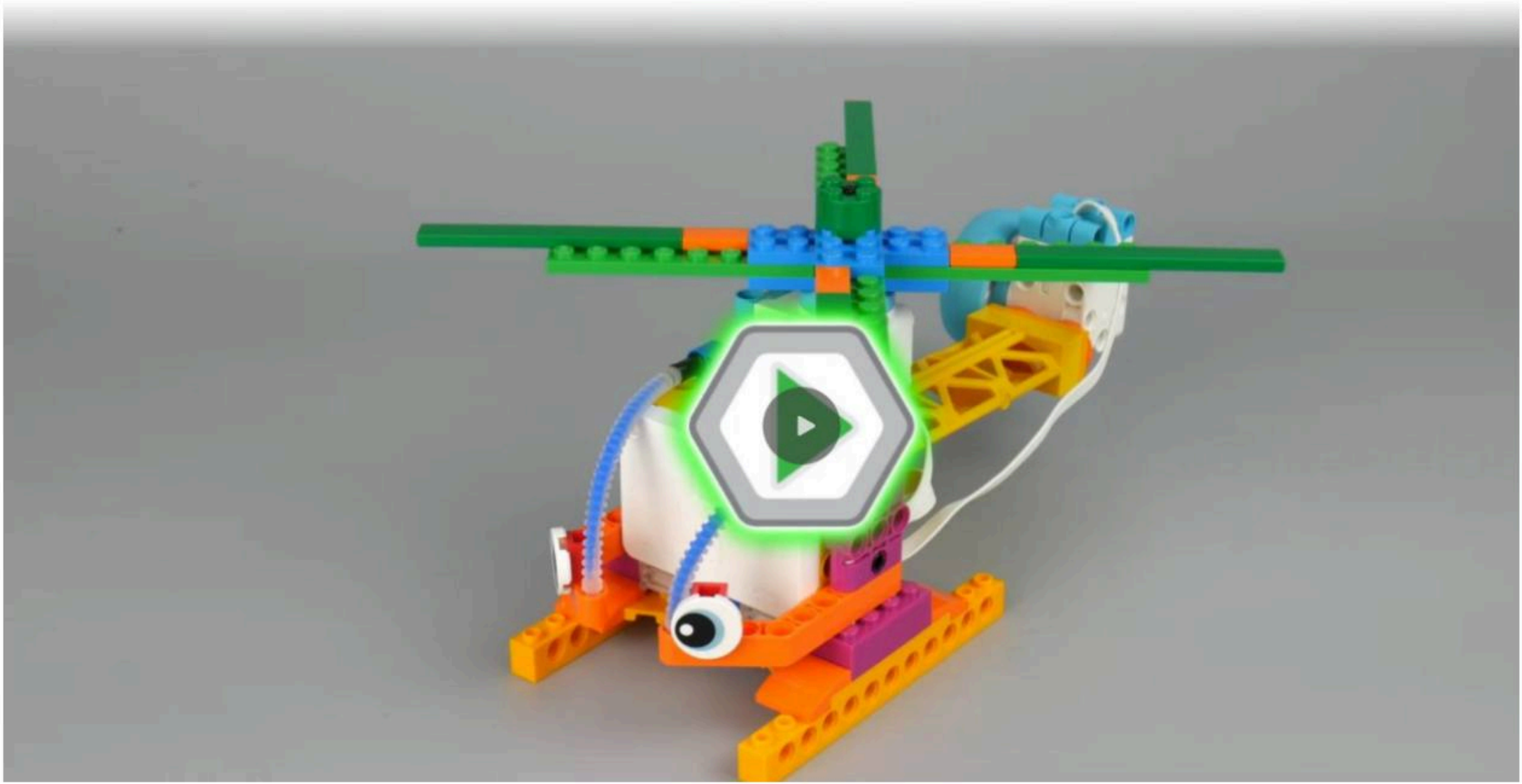








Add the sound of a flying helicopter to your playback program. In order for the sound to play continuously, use a loop



Loop

In programming, a loop is a structure that allows you to repeat a part of a program.

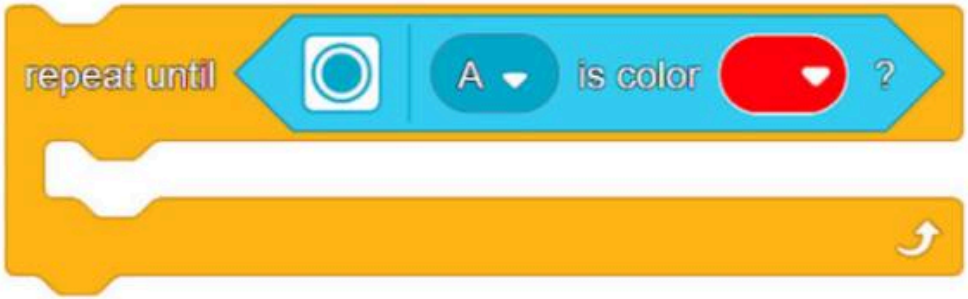
The loop can be repeated a predetermined number of times, be infinite, or there are cycles with an exit from them when a certain condition occurs



an endless loop



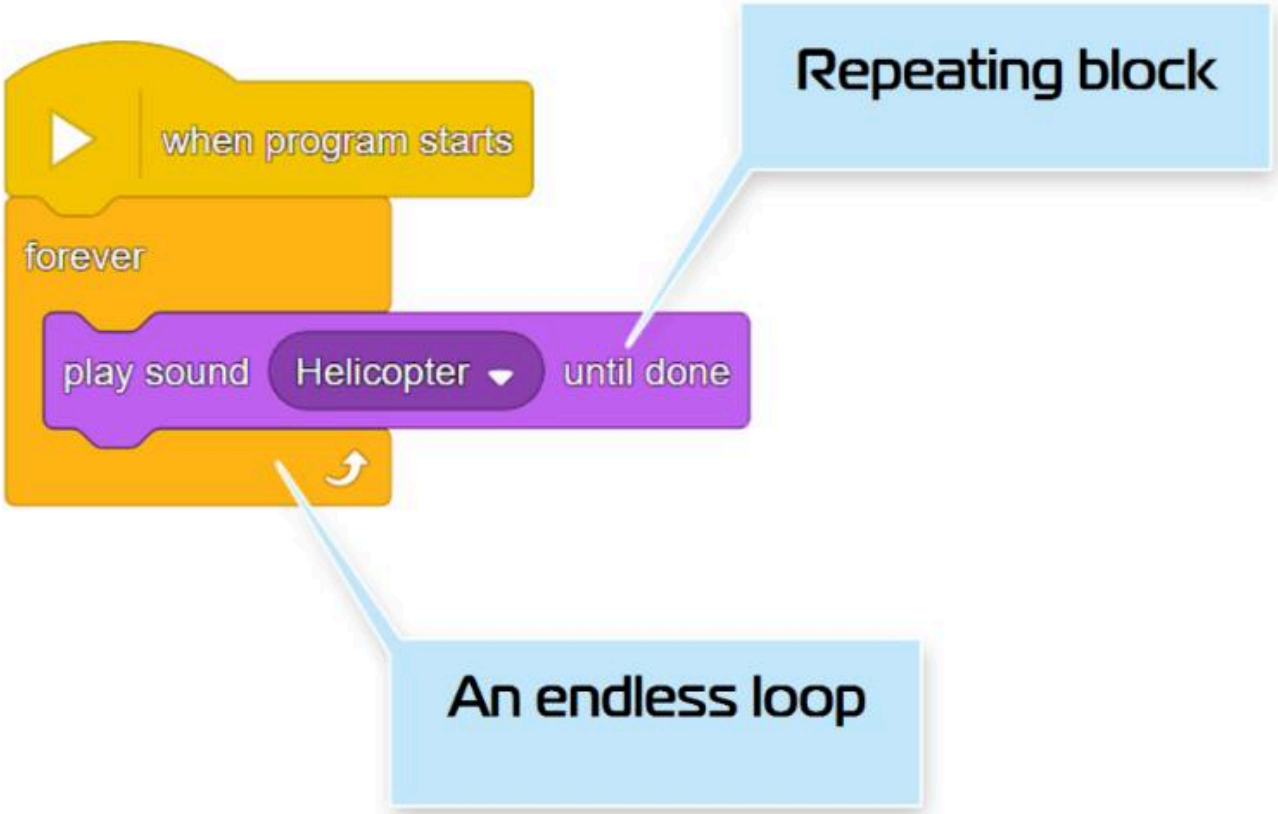
loop for 10 repetitions



the program exits the loop when red is detected

Loop

We will use an infinite loop. It will repeat the execution of the block we will place inside it:

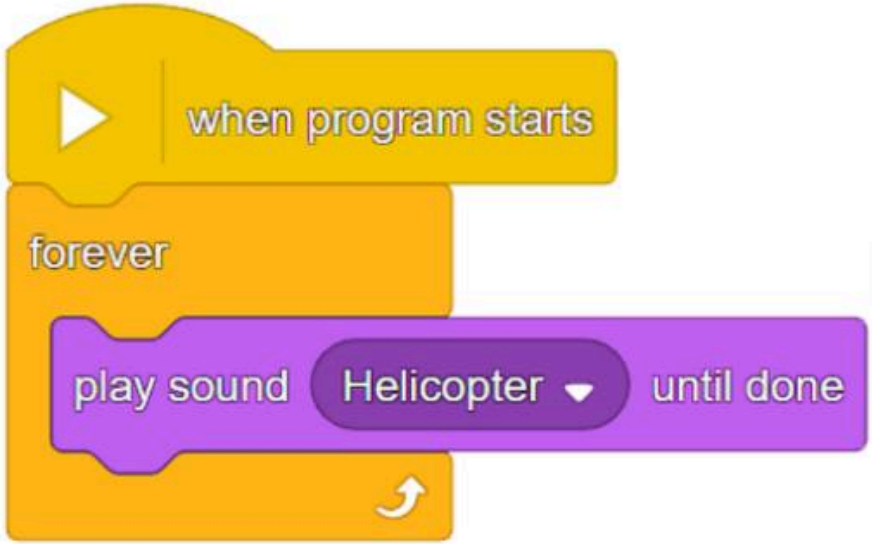


Task 6. Program

ICON BLOCKS | WORD BLOCKS

Program and test your robot.

In Word Blocks mode, the program looks like this:



Fly over the caldera and place sensors that will help track the rate of surface ascent.



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What is the purpose of the tail screw?

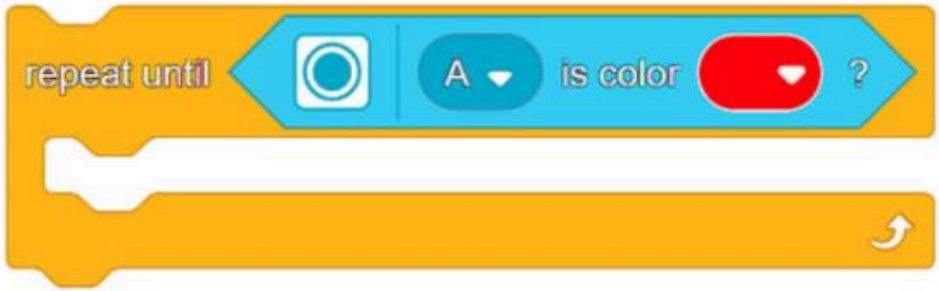


- Lifting the helicopter
- Pushing the helicopter straight forward
- Compensating the main rotor turning moment

Select the Loop block that will execute the inner blocks exactly the specified number of times.







Discuss!

- What's wrong with Sean's friend, Ashley?
- How are supervolcanoes formed?
- How often do supervolcano eruptions occur?
- Why can the eruption of a supervolcano be dangerous for the whole planet?
- What makes a helicopter better than a plane for your task?
- What sensors of the robot allow to detect its position in space?
- What block is used to repeat part of a program?



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